

Classical Forensic Feature-Based Deepfake Image Detection Using PCA and SVM

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**Classical Forensic Feature-Based Deepfake Image Detection Using PCA and SVM**” is the bonafide work of “**Madhar Husain M (24PGFR011)**” who carried out the project work under my supervision.

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ABSTRACT

The rapid advancement of artificial intelligence has enabled the creation of highly realistic synthetic media known as deepfakes. Deepfake images are generated or manipulated using advanced machine learning techniques, making it increasingly difficult to distinguish authentic images from forged ones through visual inspection alone. The misuse of such manipulated media poses significant threats to digital security, social trust, and information integrity. Therefore, the development of reliable detection mechanisms has become an important area of research in digital image forensics. This study proposes a traditional machine learning–based approach for detecting deepfake images using forensic feature analysis. Error Level Analysis (ELA) is used to reveal inconsistencies in JPEG compression that may indicate image manipulation. From the ELA-transformed images, statistical features are extracted to capture artifacts associated with deepfake generation, and Principal Component Analysis (PCA) is applied to reduce the dimensionality of the feature space while preserving the most important information.

A Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel is used to classify images as real or fake due to its effectiveness in handling high-dimensional feature spaces and nonlinear decision boundaries. The dataset used in this study consists of a balanced collection of real and manipulated images, and model evaluation is performed using a stratified 80:20 train–test split along with five-fold cross-validation. The performance of the proposed system is measured using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC). Experimental results show that the model achieves an accuracy of approximately 67%, demonstrating that traditional machine learning techniques combined with forensic feature extraction can provide a computationally efficient and interpretable approach for deepfake image detection.

Keywords: Deepfake Detection, Error Level Analysis, Machine Learning, Principal Component Analysis, Support Vector Machine, Digital Image Forensics.

CHAPTER 1

INTRODUCTION

In the modern digital era, the creation, sharing, and consumption of multimedia content have increased significantly due to the rapid growth of internet technologies and digital communication platforms. Digital images and videos are widely used in various domains such as social media, journalism, entertainment, education, and online communication. With the widespread availability of smartphones, digital cameras, and high-speed internet connectivity, people can easily capture, edit, and distribute multimedia content across the world within seconds. As a result, digital media has become an essential component of modern information exchange and communication [1].

Along with the growth of digital media technologies, advancements in artificial intelligence (AI) and machine learning have greatly enhanced the capabilities of multimedia processing. Artificial intelligence enables computers to analyze, interpret, and generate complex data such as images, audio, and video. Modern AI algorithms can perform sophisticated tasks including facial recognition, image enhancement, automatic video editing, and content generation. These technologies have transformed many industries by improving efficiency, automation, and user experience in digital media applications [2].

One of the most notable developments resulting from recent AI advancements is the emergence of synthetic media technologies. Synthetic media refers to digitally generated or manipulated content created using artificial intelligence algorithms. Among these technologies, deepfake generation has gained significant attention due to its ability to produce highly realistic manipulated images and videos. Deepfake technology typically uses advanced machine learning models to replace faces, alter expressions, or synthesize entirely new visual content that closely resembles real individuals. These manipulated media files often appear authentic to human observers, making them difficult to identify without specialized detection techniques [3].

Deepfake images are artificially generated or manipulated images created using advanced Artificial Intelligence and deep learning techniques. These images appear highly realistic and are often difficult for humans to identify as fake. The rapid growth of deepfake technology has raised serious concerns in digital media because it can be misused for identity fraud, misinformation, and digital manipulation (Rossler et al., 2019). Therefore, detecting manipulated images has become an important task in digital forensics to ensure the authenticity and reliability of digital evidence (Li & Lyu, 2019). In this study, a classical forensic feature-based approach using Error Level Analysis (ELA), Principal Component Analysis (PCA), and Support Vector Machine (SVM) is used to detect deepfake images[4].

1.1 Digital Media and the Growth of Artificial Intelligence

Digital media has become an integral part of modern communication and information exchange. With the rapid development of the internet, smartphones, and high-speed data networks, digital images and videos are now widely used for personal, educational, commercial, and social purposes. Platforms such as social media networks, online news portals, and multimedia sharing websites allow users to create and distribute digital content instantly across the globe. As a result, digital images have become one of the most common forms of communication in today's digital environment [5].

Over the past two decades, the advancement of artificial intelligence (AI) has significantly transformed the way digital media is created, processed, and analyzed. Artificial intelligence refers to the development of computer systems that can perform tasks typically requiring human intelligence, such as pattern recognition, decision making, and problem solving. Machine learning, a major branch of AI, enables computers to learn patterns from data and improve their performance over time without being explicitly programmed for every task. These capabilities have enabled new technologies that can automatically edit, enhance, and generate digital images with remarkable realism [6].

The integration of AI with digital media processing has led to major innovations in image editing, computer vision, and multimedia generation. Modern AI algorithms can automatically enhance image quality, remove unwanted objects, generate artistic effects, and even create entirely new images that appear highly realistic. These technologies are widely used in industries such as entertainment, advertising, healthcare, gaming, and virtual reality. For example, AI-powered tools are commonly used in photo editing applications to improve image resolution, adjust lighting conditions, and perform complex visual transformations [7].

1.2 Emergence of Synthetic Media and Deepfakes

The rapid progress of artificial intelligence and machine learning technologies has significantly transformed the field of digital media creation. In recent years, a new category of media known as **synthetic media** has emerged. Synthetic media refers to digital content

such as images, videos, or audio that is either partially or entirely generated or modified using artificial intelligence algorithms. These technologies allow computers to produce highly realistic visual and audio content that can closely resemble real-world data. As a result, the line between authentic media and artificially generated content has become increasingly difficult to distinguish [8].

One of the most widely discussed forms of synthetic media is deepfake technology. The term “deepfake” is derived from the combination of “deep learning” and “fake,” reflecting the use of deep learning techniques to generate manipulated media. Deepfake technology gained widespread attention around 2017 when manipulated videos began appearing on online platforms, demonstrating the ability to realistically replace one person's face with another in video content. Since then, deepfake technology has rapidly evolved, enabling the generation of highly convincing synthetic images, videos, and audio recordings [9].

Deepfake generation techniques typically rely on advanced machine learning models that learn patterns from large datasets of images and videos. By analyzing facial structures, expressions, and movements, these models can generate synthetic representations that mimic real individuals. One of the common techniques used in early deepfake systems involved **autoencoders**, which are neural network models designed to learn compressed representations of data and reconstruct them. In face-swapping applications, autoencoders can learn to encode facial features of one individual and decode them onto another face, creating a realistic transformation [10].

Another major technological advancement contributing to deepfake generation is the development of **Generative Adversarial Networks (GANs)**. GANs consist of two neural networks known as the generator and the discriminator. The generator attempts to produce synthetic images that resemble real data, while the discriminator attempts to distinguish between real and generated images. Through continuous competition during training, the generator gradually improves its ability to produce highly realistic images that can be difficult to differentiate from genuine photographs. This adversarial learning process has significantly improved the quality of synthetic media generation [11].

As deepfake technology has improved, the quality and realism of generated images and videos have increased substantially. Modern deepfake tools can replicate facial

expressions, lighting conditions, and fine visual details with high accuracy. Furthermore, many user-friendly deepfake software tools have become publicly available, allowing individuals with limited technical knowledge to create manipulated media content. The accessibility of these tools has accelerated the spread of synthetic media across various online platforms [12].

1.3 Working Principles of Deepfake Generation

Deepfake technology is based on advanced artificial intelligence techniques that enable computers to generate or manipulate digital images and videos in a highly realistic manner. The primary objective of deepfake generation is to synthesize visual content that closely resembles real individuals, often by replacing faces, modifying expressions, or generating entirely new identities. These manipulations are achieved using machine learning models that learn patterns from large datasets of images or videos. By analyzing facial structures, movements, and visual features, these models can produce synthetic media that is difficult to distinguish from authentic content [13].

One of the earliest approaches used in deepfake generation involves autoencoder-based architectures. An autoencoder is a type of neural network designed to learn compressed representations of data and reconstruct the original input from that representation. In the context of deepfake creation, autoencoders are trained on images of different individuals. The encoder part of the network learns to capture important facial features, while the decoder reconstructs the face using this learned representation. By training separate decoders for different individuals while sharing the same encoder, the system can effectively swap faces between people. This technique allows the facial structure of one individual to be mapped onto another person's face in images or video frames [14].

Another widely used technique for deepfake generation is based on Generative Adversarial Networks (GANs). GANs consist of two neural networks that work in opposition to each other during the training process. The first network, known as the generator, is responsible for creating synthetic images from random noise or learned representations. The second network, called the discriminator, attempts to determine whether an image is real or generated. During training, the generator continuously improves its ability to produce realistic images in order to deceive the discriminator, while the discriminator becomes

better at identifying fake images. This adversarial training process results in highly realistic synthetic images that closely resemble real-world photographs [15].

In addition to GAN-based methods, face swapping techniques are commonly used in deepfake generation. Face swapping involves detecting and extracting facial regions from images or video frames and replacing them with another person's face while maintaining natural alignment and facial expressions. This process typically includes several steps such as face detection, facial landmark identification, alignment, blending, and color correction. By carefully adjusting these elements, the manipulated image can appear natural and visually convincing [16].

Another important aspect of deepfake generation is the replication of facial expressions and movements. Advanced deepfake systems analyze facial landmarks such as the eyes, nose, mouth, and jawline to track facial movements in videos. These movements are then transferred to the synthesized face to maintain realistic expressions and lip synchronization. This capability allows deepfake videos to mimic speech patterns and emotional expressions with high accuracy, further enhancing their realism [17].

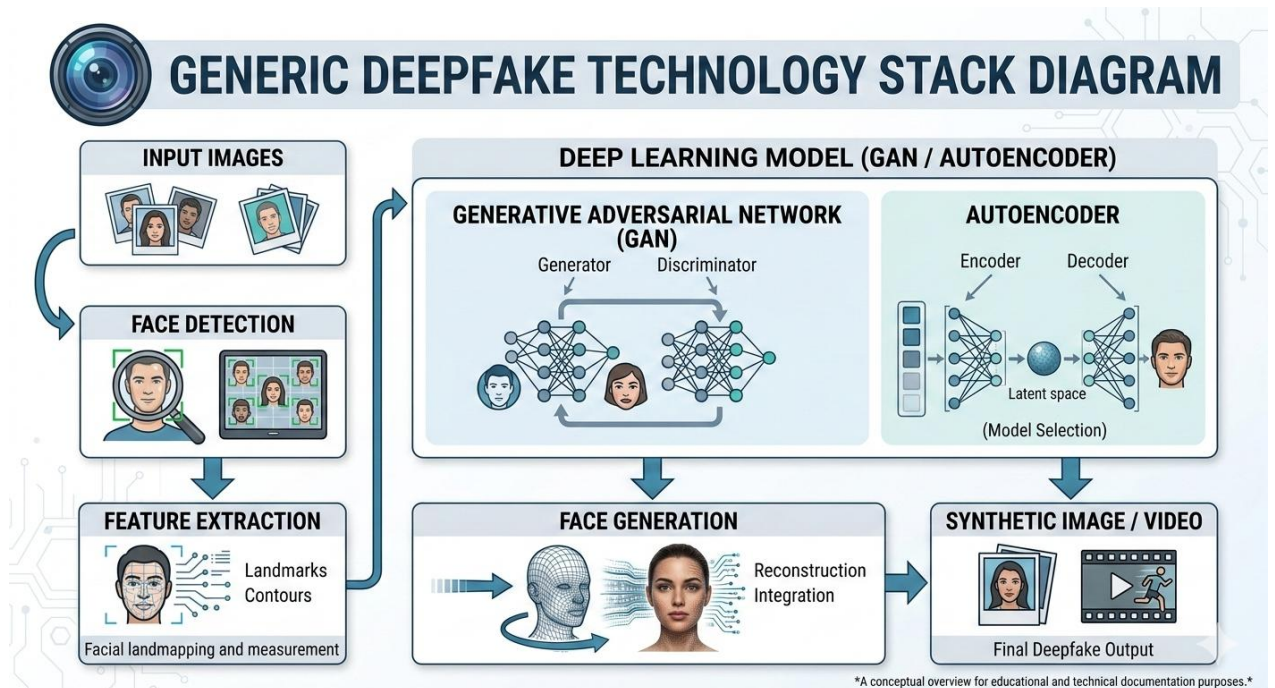


Figure 1.1: General process of deepfake generation.

1.4 Applications of Deepfake Technology

Deepfake technology, although widely associated with misinformation and manipulation, also has several legitimate and beneficial applications across different industries. The ability of artificial intelligence to generate realistic synthetic images and videos has opened new possibilities in areas such as entertainment, education, virtual communication, and digital content creation. When used responsibly, deepfake technology can enhance creativity, reduce production costs, and enable new forms of interactive media experiences [18].

One of the most prominent applications of deepfake technology can be found in the entertainment and film industry. Film producers and visual effects artists use AI-based techniques to modify or recreate actors' appearances in movies. For example, deepfake technology can be used to digitally recreate younger versions of actors, replace stunt performers with the actor's face, or restore the appearance of historical characters. These techniques allow filmmakers to produce realistic visual effects without requiring complex manual editing. In some cases, deepfake technology can even recreate performances of actors who are no longer alive, enabling the continuation of certain film projects or historical reconstructions [19].

Another important application of deepfake technology is in the field of digital content creation and media production. Content creators and video editors can use AI-based tools to modify facial expressions, synchronize lip movements with translated speech, or generate realistic avatars for videos. This capability is particularly useful in international media production where content needs to be adapted for different languages. AI-based lip synchronization allows the speaker's mouth movements to match translated audio, improving the viewing experience for audiences across different regions [20].

Deepfake technology is also being explored in education and training environments. Educational institutions and training organizations can use synthetic media to create interactive learning experiences. For example, historical figures can be digitally recreated to present lectures or narrate historical events, making educational content more engaging for students. In professional training programs, simulated characters can be used to demonstrate real-world scenarios such as medical procedures, emergency response situations, or customer service interactions [21].

In addition to education, deepfake technology plays an important role in the development of virtual assistants and digital avatars. Many modern applications involve the use of AI-generated avatars to interact with users in virtual environments. These avatars can replicate human-like facial expressions and speech patterns, providing a more natural communication experience. Virtual presenters and AI-generated spokespersons are increasingly used in online customer support systems, marketing campaigns, and digital presentations [22].

1.5 Threats and Misuse of Deepfakes

While deepfake technology offers several beneficial applications, its rapid advancement has also introduced serious risks and ethical concerns. The ability to create highly realistic manipulated images and videos makes deepfake technology a powerful tool that can be exploited for malicious purposes. As deepfake generation tools become more accessible and easier to use, the potential for misuse increases significantly. This has raised concerns among researchers, policymakers, and cybersecurity professionals regarding the impact of deepfakes on digital trust and information security [23].

One of the most significant threats associated with deepfakes is the spread of misinformation and disinformation. Deepfake media can be used to fabricate events or statements that never actually occurred, making it difficult for viewers to distinguish between genuine and manipulated content. Such fabricated media can be used to mislead the public, distort facts, or influence public opinion. In the context of political communication, manipulated videos or images of public figures can potentially damage reputations, spread false narratives, and create confusion among audiences. The rapid dissemination of such content through social media platforms further amplifies its impact [24].

Another major concern is identity misuse and impersonation. Deepfake technology can be used to create realistic images or videos that impersonate individuals without their consent. This form of digital impersonation can lead to privacy violations and reputational damage. For example, manipulated media may falsely depict individuals in situations that never occurred, potentially harming their personal or professional lives

In some cases, deepfakes have been used to create fraudulent content involving celebrities, public figures, or private individuals[25].

Deepfake technology also poses a threat in the area of cybercrime and financial fraud. Cybercriminals may use synthetic media to impersonate company executives, business partners, or trusted individuals in order to manipulate victims into transferring money or sensitive information. In certain cases, deepfake audio or video recordings have been used in social engineering attacks where attackers attempt to gain trust by appearing as legitimate authorities. These sophisticated impersonation techniques can make fraudulent activities more convincing and difficult to detect [26].

Another critical concern is the potential use of deepfake technology for harassment and defamation. Manipulated images or videos may be used to spread false allegations, humiliate individuals, or damage reputations. Victims of such attacks may face emotional distress, social consequences, and legal complications. Since deepfake media can appear highly realistic, it can be challenging for individuals to prove that the content is fabricated, especially when it spreads rapidly across online platforms [26].

1.6 Importance of Deepfake Detection

The rapid advancement of deepfake generation technologies has created a growing need for reliable methods to identify manipulated digital media. As synthetic images and videos become increasingly realistic, distinguishing between authentic and fabricated content has become a major challenge. Deepfake detection plays a crucial role in preserving the authenticity of digital information and protecting individuals, organizations, and society from the harmful consequences of manipulated media [27].

One of the primary reasons deepfake detection is important is the protection of information integrity. In today's digital environment, people often rely on images and videos as evidence of real-world events. News media, social media platforms, and online communication channels frequently distribute visual content that influences public perception. If manipulated media spreads without proper verification, it can lead to the dissemination of false information and misinterpretation of events. Effective deepfake detection systems help ensure that digital media remains a trustworthy source of information [28].

Another significant reason for deepfake detection is the protection of individual privacy and identity. Deepfake technology can be used to impersonate individuals without their consent, creating synthetic media that falsely represents them in various situations. Such impersonation can lead to reputational damage, harassment, and privacy violations. Detection techniques can help identify manipulated media early and prevent the misuse of an individual's identity in digital environments [29].

Deepfake detection is also important in the context of cybersecurity and fraud prevention. Malicious actors may use deepfake images or videos to deceive individuals, organizations, or financial institutions. For example, synthetic media can be used in social engineering attacks where attackers impersonate trusted individuals to gain unauthorized access to sensitive information or financial resources. By developing systems capable of identifying manipulated content, organizations can reduce the risk of such fraudulent activities [30].

Another critical area where deepfake detection is essential is digital forensics and law enforcement. Digital images and videos are often used as evidence in criminal investigations, legal proceedings, and forensic analysis. The presence of deepfake technology introduces new challenges in verifying the authenticity of such evidence. Investigators must be able to determine whether digital media has been altered or artificially generated. Deepfake detection techniques provide forensic analysts with tools to analyze visual artifacts and confirm the authenticity of digital content [31].

1.7 Overview of Digital Image Forensics

Digital image forensics is a specialized field within digital forensics that focuses on analyzing digital images to determine their authenticity and identify any form of manipulation. With the widespread use of digital cameras, smartphones, and image editing software, digital images can easily be altered or fabricated. As a result, verifying the integrity and originality of images has become an important task in areas such as cybersecurity, law enforcement, journalism, and legal investigations. Digital image

forensics provides techniques and tools that help detect whether an image has been modified, manipulated, or artificially generated [32].

The primary objective of digital image forensics is to examine images for evidence of tampering or manipulation. This process involves analyzing various characteristics of an image, including its metadata, compression artifacts, pixel structure, and statistical patterns. By carefully examining these characteristics, forensic analysts can identify inconsistencies that may indicate that the image has been altered. Such inconsistencies may arise due to editing operations such as cropping, splicing, retouching, or synthetic content generation.

Digital image forensics techniques are generally classified into two main categories: active forensics and passive forensics. Active forensics methods involve embedding additional information into the image at the time of creation. This information, often referred to as a digital watermark or signature, can later be used to verify the authenticity of the image. If the embedded information is missing or altered, it indicates that the image may have been tampered with. However, active forensic methods require special hardware or software during the image capture process, which limits their practical use in many real-world situations [33].

In contrast, passive forensics, also known as blind image forensics, does not require any prior embedded information in the image. Instead, it relies on analyzing intrinsic properties and statistical patterns present within the image itself. Passive forensic techniques examine artifacts introduced during image acquisition, compression, or editing. These artifacts may include irregular compression levels, inconsistent lighting patterns, abnormal pixel correlations, or unnatural texture distributions. Since most images available on the internet do not contain embedded watermarks, passive forensic techniques are widely used for detecting image manipulation [34].

1.8 Types of Image Manipulation Techniques

Digital image manipulation refers to the process of altering the content or appearance of an image using various editing techniques or software tools. With the advancement of image editing applications and artificial intelligence technologies, modifying digital images has become easier and more sophisticated. Image manipulation can be performed

for legitimate purposes such as improving visual quality or creating artistic effects. However, it can also be used to produce misleading or fabricated content. Understanding the different types of image manipulation techniques is important for developing effective methods to detect altered images [35].

One common form of image manipulation is image splicing. Image splicing involves combining elements from two or more different images to create a new composite image. In this process, specific objects or regions are extracted from one image and inserted into another image. The goal is usually to make the final image appear realistic and seamless. However, splicing may introduce inconsistencies in lighting, shadows, color balance, or texture patterns, which can sometimes be detected through forensic analysis. Image splicing is widely used in digital editing and can also be exploited to fabricate misleading visual content [36].

Another widely used manipulation technique is copy-move forgery. In copy-move manipulation, a portion of an image is copied and pasted onto another location within the same image. This technique is often used to hide or duplicate objects within an image. For example, unwanted elements may be concealed by copying nearby background regions and placing them over the target area. Because the copied region originates from the same image, the color and texture patterns often appear consistent, making this type of manipulation more difficult to detect [37].

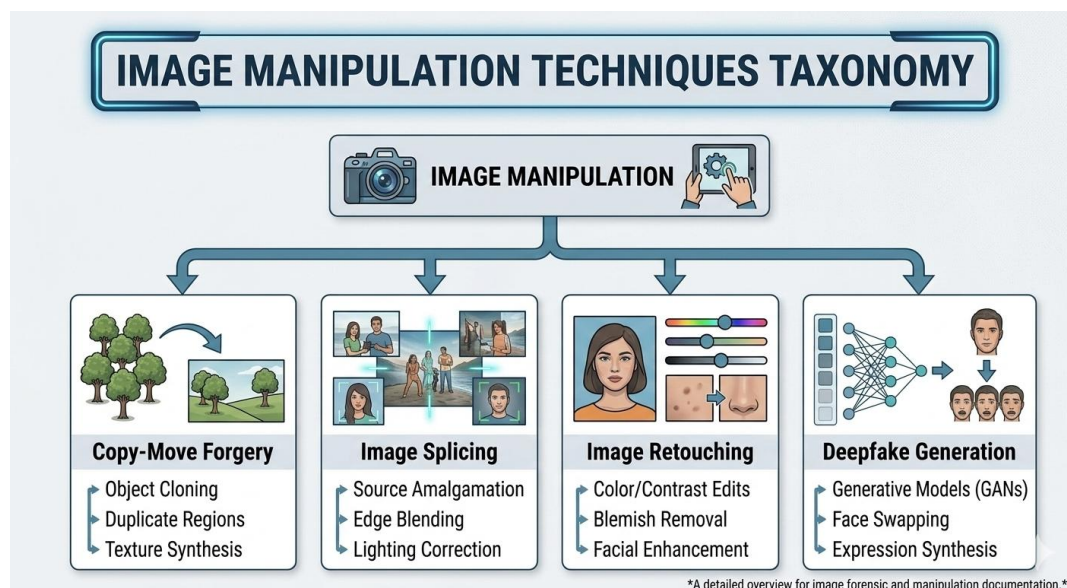


Figure 1.2: Classification of Common Image Manipulation Techniques.

1.9 Existing Deepfake Detection Approaches

With the rapid development of deepfake generation technologies, researchers have proposed various methods to detect manipulated images and videos. Detecting deepfakes is a challenging task because modern synthetic media generation techniques can produce highly realistic visual content that closely resembles genuine media. To address this challenge, different computational approaches have been developed to analyze subtle artifacts and inconsistencies introduced during the deepfake creation process. These

detection methods can generally be categorized into two major groups: deep learning-based approaches and traditional machine learning-based approaches[38].

One of the most widely used approaches for deepfake detection involves deep learning techniques, particularly convolutional neural networks (CNNs). Deep learning models are capable of automatically learning complex patterns and visual features directly from raw image data. In deepfake detection, CNN-based architectures are trained on large datasets containing both real and manipulated images. During training, the neural network learns to identify subtle visual cues that distinguish authentic images from synthetic ones. These cues may include abnormal facial textures, irregular lighting patterns, or inconsistencies in image compression artifacts. Deep learning methods have demonstrated strong performance in many deepfake detection studies due to their ability to capture complex visual patterns [39].

In addition to CNNs, researchers have also explored other deep learning models such as recurrent neural networks (RNNs) and transformer-based architectures for analyzing video-based deepfakes. These models are particularly useful for detecting temporal inconsistencies across video frames, such as unnatural facial movements or irregular blinking patterns. By analyzing sequences of frames rather than individual images, these methods can identify anomalies that occur during deepfake video generation [40].

Despite their high accuracy, deep learning-based detection methods have several limitations. Training deep neural networks typically requires very large datasets, powerful hardware resources such as graphics processing units (GPUs), and significant computational time. Additionally, deep learning models often function as “black-box” systems, meaning that their decision-making processes can be difficult to interpret. These

limitations can make deep learning methods less suitable for applications that require transparency, lower computational cost, or deployment on resource-constrained systems [41].

1.10 Forensic Significance

Forensic significance refers to the role of forensic analysis in verifying the authenticity and integrity of digital images. With the increasing use of digital media, images can be easily edited or manipulated using advanced software tools. Such manipulations may be difficult to detect with the human eye [42].

Digital image forensics helps identify hidden traces of manipulation by analyzing image properties such as compression artifacts, pixel inconsistencies, and statistical patterns. Techniques like Error Level Analysis (ELA), feature extraction, and machine learning algorithms are widely used to detect these irregularities [43].

In the context of deepfake detection, forensic analysis is highly significant because it helps distinguish between real and manipulated images. By applying forensic techniques, it becomes possible to identify forged content and prevent the spread of misleading or false information. Therefore, forensic methods play an important role in improving the reliability and trustworthiness of digital media [44].

CHAPTER 2

REVIEW OF LITERATURE

2.1 Existing Research

Several research studies have been conducted in recent years to address the growing challenge of detecting deepfake images and videos. These studies have explored a wide range of techniques including traditional machine learning methods, deep learning architectures, and hybrid approaches that combine different feature extraction strategies. Reviewing these studies provides valuable insights into the strengths and limitations of existing deepfake detection methods.

Research Papers & Journal Papers

Bharati et al. (2025) introduced a study titled “*Explainable Deepfake Detection: A Multi-Model Framework*,” published in the ACM Digital Library. The authors proposed an explainable artificial intelligence framework for detecting manipulated media. Their approach integrates multiple detection models while providing explanations for classification decisions. This improves transparency in deepfake detection systems used in forensic investigations. The research contributes to the development of trustworthy and interpretable detection models.

Heidari et al. (2024) presented a paper titled “*Deepfake Detection using Deep Learning Methods: A Review*,” published in Wiley Interdisciplinary Reviews. The study provides a detailed review of deep learning architectures used in deepfake detection research. The authors discussed convolutional neural networks, transformer models, and hybrid detection techniques. They also analyzed datasets, evaluation metrics, and challenges associated with deepfake detection. The paper serves as a comprehensive reference for researchers working in this field.

Karaköse et al. (2024) conducted a study titled “*A New Approach for Deepfake Detection with Choquet Fuzzy Integral*,” published in Applied Sciences (MDPI). The researchers proposed an ensemble learning technique that integrates multiple classifiers using a fuzzy integral method. This approach improves the reliability of detection by combining predictions from different models. The study demonstrated improved accuracy compared with traditional machine learning methods. The research highlights the benefits of ensemble techniques in deepfake detection.

Zhang et al. (2024) presented a research paper titled *“TSFF-Net: DeepFake Video Detection Model,”* published in the journal Scientific Reports. The authors developed a deep learning architecture based on multi-scale feature fusion. The proposed model analyzes both spatial and temporal patterns present in manipulated videos. By combining multiple feature representations, the model improves detection performance. Experimental results showed strong accuracy on commonly used deepfake datasets.

Naskar et al. (2024) proposed a study titled *“Deepfake Detection using Deep Feature Stacking and Meta-Learning,”* published in IEEE Access. The research introduced a feature stacking approach that combines multiple deep learning models to improve detection accuracy. The method integrates features from several neural networks into a meta-learning framework. This combination improves the ability of the model to capture subtle manipulation artifacts. The results demonstrated improved detection accuracy compared with single-model approaches.

Akhtar et al. (2023) presented a paper titled *“Deepfakes Generation and Detection: A Short Survey,”* published in IEEE Access. The study reviewed different techniques used to generate deepfake images and videos using artificial intelligence. It also analyzed existing detection methods designed to identify manipulated content. The authors discussed challenges related to high-quality deepfake generation and the limitations of current detection approaches. The survey provided an overview of emerging technologies used in deepfake research.

Gupta et al. (2023) published a comprehensive study titled *“A Comprehensive Review of DeepFake Detection Using Machine Learning,”* in the journal Electronics (MDPI). The paper provided an extensive survey of deepfake detection methods developed in recent years. The authors discussed machine learning and deep learning techniques used to detect manipulated media. They also examined available datasets and evaluation metrics used in the research community. The study helped researchers understand current challenges and future directions in deepfake detection.

Alanazi et al. (2023) presented research titled *“Improving Detection of DeepFakes through Facial Region Analysis,”* published in the journal Electronics (MDPI). The authors

proposed a deepfake detection technique focusing on specific facial regions such as eyes, mouth, and nose. These areas were analyzed individually to identify inconsistencies caused by image manipulation. Machine learning models were trained on extracted facial region features. The results showed that analyzing critical facial areas improved detection performance.

Khan et al. (2023) conducted a study titled “*Deepfake Detection: A Comparative Analysis*,” published in IEEE Access. The researchers compared several machine learning and deep learning techniques used for detecting manipulated media. Their analysis included convolutional neural networks, transformer architectures, and hybrid detection models. The study evaluated performance across multiple datasets to identify the strengths and weaknesses of each method. The results provided valuable insights into the effectiveness of different deepfake detection approaches.

Vamsi et al. (2022) published a study titled “*Deepfake Detection in Digital Media Forensics*,” in the Journal of Information Security. The authors investigated the use of deep learning architectures for identifying manipulated images and videos. Their model combined convolutional neural networks with long short-term memory networks to capture spatial and temporal patterns. The proposed method analyzed facial movements and frame consistency in videos. Experimental results indicated improved performance in detecting deepfake media compared with traditional detection methods.

Tolosana et al. (2022) presented a research study titled “*DeepFake Detection in Digital Media Forensics*,” published in the journal Applied Soft Computing. The paper focused on detecting manipulated media by analyzing facial features and spatial patterns. The authors proposed a detection approach that analyzes facial regions individually to identify manipulation artifacts. Their method used convolutional neural networks for feature extraction and classification. The study provided an overview of deepfake detection challenges and demonstrated improved detection accuracy across multiple datasets.

Ramachandran et al. (2021) conducted research titled “*An Experimental Evaluation on Deepfake Detection using Deep Face Recognition*,” published in the IEEE International Conference on Biometrics. The study explored the use of deep face recognition networks for detecting manipulated media. The authors evaluated several facial recognition

architectures on deepfake datasets such as FaceForensics++. Their approach analyzed facial feature inconsistencies caused by manipulation techniques. The results showed high detection accuracy, demonstrating the potential of deep face recognition models for deepfake detection tasks.

Nirkin et al. (2020) introduced a study titled *“DeepFake Detection Based on the Discrepancy Between the Face and its Context,”* published in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). The researchers proposed a detection method that analyzes inconsistencies between the manipulated facial region and the surrounding context of the image. They utilized two neural networks to separately analyze the face and the background region. By comparing these regions, the model could identify anomalies introduced during deepfake generation. The proposed method achieved strong detection accuracy and demonstrated the importance of contextual information in deepfake detection.

Li et al. (2020) proposed a research work titled *“Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics,”* published in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). The authors developed a new dataset containing highly realistic deepfake videos created using advanced synthesis techniques. The dataset was designed to address limitations present in earlier datasets that contained visible artifacts. The researchers evaluated multiple deep learning detection models on this dataset. Their results demonstrated the increasing difficulty of detecting high-quality deepfakes and highlighted the need for improved detection algorithms.

Verdoliva (2020) published an influential paper titled *“Media Forensics and DeepFakes: An Overview,”* in the IEEE Journal of Selected Topics in Signal Processing. The study provides a detailed overview of deepfake generation techniques and detection strategies. It discusses both traditional forensic analysis methods and modern deep learning approaches. The paper also explores challenges related to adversarial attacks and evolving manipulation techniques. The research has become a key reference for understanding the field of deepfake detection.

Mittal et al. (2020) conducted a study titled *“A Survey of Deepfake Detection Techniques,”* published in the Journal of Artificial Intelligence Research. The paper provides a

comprehensive overview of different deepfake detection techniques developed in recent years. The authors reviewed machine learning, deep learning, and hybrid detection approaches. They also analyzed available datasets and evaluation metrics used in deepfake research. The study highlights the challenges associated with detecting highly realistic synthetic media.

Wang et al. (2020) published a study titled “*CNN-generated Images Are Surprisingly Easy to Spot... for Now,*” in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). The researchers analyzed patterns present in images generated by generative adversarial networks. They discovered that synthetic images often contain statistical inconsistencies that can be detected using machine learning algorithms. The study proposed a forensic detection method based on frequency artifacts. Their work demonstrated that GAN-generated images leave identifiable signatures.

Dolhansky et al. (2020) introduced a research work titled “*The DeepFake Detection Challenge Dataset,*” presented in the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). The study describes the development of a large-scale dataset created for the DeepFake Detection Challenge. The dataset contains thousands of manipulated videos generated using different deepfake techniques. Researchers evaluated several deep learning models using this dataset. The work has significantly contributed to the development of improved deepfake detection systems.

Rössler et al. (2019) presented a study titled “*FaceForensics++: Learning to Detect Manipulated Facial Images,*” published in the IEEE International Conference on Computer Vision (ICCV). The authors introduced one of the largest publicly available datasets for deepfake detection research. The dataset contains more than 1.8 million manipulated images generated using multiple face manipulation techniques. They evaluated several convolutional neural network models for detecting manipulated facial images. Their work significantly contributed to deepfake research by providing a benchmark dataset widely used for evaluating detection algorithms.

Nguyen et al. (2019) presented a study titled “*Capsule-Forensics: Using Capsule Networks to Detect Forged Images and Videos,*” published in the IEEE International

Conference on Acoustics, Speech and Signal Processing (ICASSP). The authors proposed a deepfake detection method based on capsule networks. Capsule networks capture spatial relationships between features more effectively than traditional convolutional networks. The method analyzes hierarchical patterns within images to identify manipulated regions. Experimental results showed that capsule networks can effectively detect both image and video forgeries.

Sabir et al. (2019) introduced a study titled “*Recurrent Convolutional Strategies for Face Manipulation Detection*,” published in the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). The authors proposed a hybrid model combining convolutional neural networks with recurrent neural networks. This architecture captures both spatial features and temporal inconsistencies in manipulated videos. The model was evaluated using the FaceForensics dataset and achieved improved detection accuracy. The research highlighted the importance of temporal modeling in detecting deepfake videos.

Guera and Delp (2018) conducted research titled “*Deepfake Video Detection Using Recurrent Neural Networks*,” presented at the IEEE International Conference on Advanced Video and Signal Based Surveillance. The authors proposed a deep learning model that combines convolutional neural networks with recurrent neural networks. Their method analyzes temporal inconsistencies across video frames to detect manipulated content. The model identifies irregular facial movements and unnatural patterns produced by deepfake generation algorithms. The study demonstrated that incorporating temporal information improves the reliability of deepfake detection.

Chollet (2017) introduced the Xception network architecture in the paper titled “*Xception: Deep Learning with Depthwise Separable Convolutions*,” published in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Although the paper focuses on image classification, the Xception architecture has been widely adopted for deepfake detection research. Many deepfake detection systems utilize this architecture due to its efficiency and strong feature extraction capabilities.

2.2 Case Study

Case Study 1 – 2016

Korshunov and Marcel (2016) conducted one of the earliest studies on deepfake detection titled “*DeepFakes: A New Threat to Face Recognition?*” published in the IEEE International Conference on Biometrics. The study analyzed how deepfake videos can bypass biometric authentication systems. The researchers evaluated deepfake videos against face recognition models to examine their vulnerability. Their findings revealed that face recognition systems can be fooled by synthetic faces generated using deep learning techniques. This study highlighted the need for robust detection methods to prevent misuse of facial recognition systems.

Case Study 2 – 2017

Chollet (2017) introduced the deep learning architecture in the paper “*Xception: Deep Learning with Depthwise Separable Convolutions*,” published in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Although primarily designed for image classification, the Xception architecture later became widely used for deepfake detection tasks. The model improves efficiency by separating spatial and channel-wise feature extraction. Researchers adopted this architecture for detecting manipulation artifacts in facial images. It has since become a benchmark model in deepfake detection research.

Case Study 3 – 2018

Afchar et al. (2018) developed a deepfake detection method titled “*MesoNet: A Compact Facial Video Forgery Detection Network*,” presented at the IEEE International Workshop on Information Forensics and Security (WIFS). The study proposed a lightweight convolutional neural network capable of detecting manipulated facial videos. The model focuses on mesoscopic image features rather than deep semantic features. This approach enables the detection of subtle artifacts created by face swapping techniques. Experimental results showed promising accuracy in identifying manipulated videos.

Case Study 4 – 2019

Rössler et al. (2019) introduced the widely used dataset in their paper “*FaceForensics++: Learning to Detect Manipulated Facial Images*,” published in the IEEE International Conference on Computer Vision (ICCV). The study created a large dataset containing

manipulated facial videos generated using multiple deepfake techniques. Researchers evaluated several deep learning detection models using this dataset. The dataset helped standardize the evaluation of deepfake detection algorithms. It is now one of the most widely used benchmarks in deepfake research.

Case Study 5 – 2020

Li et al. (2020) proposed the dataset “*Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics*,” published in the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). The authors created a dataset containing highly realistic deepfake videos with fewer visual artifacts. This dataset was designed to address limitations present in earlier datasets. Several detection models were evaluated using this dataset to analyze performance. The study demonstrated that detecting high-quality deepfakes remains a challenging problem.

Case Study 6 – 2021

Ramachandran et al. (2021) conducted a study titled “*An Experimental Evaluation on Deepfake Detection using Deep Face Recognition*,” published in the IEEE International Conference on Biometrics. The research explored the use of facial recognition architectures for detecting manipulated images and videos. The model analyzed inconsistencies in facial features caused by deepfake generation algorithms. Experiments were conducted using FaceForensics datasets. The results demonstrated improved detection accuracy using deep face recognition models.

Case Study 7 – 2022

Tolosana et al. (2022) published a study titled “*DeepFake Detection in Digital Media Forensics*,” in the journal Applied Soft Computing. The research analyzed different techniques used to detect manipulated media. The authors proposed detection methods that focus on facial region inconsistencies and spatial artifacts. Their approach utilized deep learning models trained on large datasets. The study demonstrated improved detection performance and highlighted challenges in detecting sophisticated deepfake media.

Case Study 8 – 2023

Khan et al. (2023) conducted a study titled “*Deepfake Detection: A Comparative Analysis*,” published in IEEE Access. The research compared multiple machine learning

and deep learning approaches used in deepfake detection. Various models such as CNNs and transformer-based architectures were evaluated across several datasets. The study analyzed detection accuracy and computational performance. The results provided insights into the strengths and weaknesses of different detection methods.

Case Study 9 – 2024

Zhang et al. (2024) proposed a model titled “*TSFF-Net: DeepFake Video Detection Model*,” published in Scientific Reports. The authors developed a deep learning architecture that integrates multi-scale feature fusion techniques. The model analyzes both spatial and temporal features present in manipulated videos. Experiments were conducted using FaceForensics++ and Celeb-DF datasets. The results showed improved detection accuracy compared with earlier deepfake detection models.

Case Study 10 – 2025

Bharati et al. (2025) introduced a study titled “*Explainable Deepfake Detection: A Multi-Model Framework*,” published in the ACM Digital Library. The research proposed an explainable artificial intelligence approach for detecting manipulated media. Multiple deep learning models were integrated into a single detection framework. The system provides interpretable explanations for classification decisions. This approach improves trust and transparency in deepfake detection systems used in digital forensics

CHAPTER 3

AIM AND OBJECTIVES

Aim of the study:

The aim of this research is to develop an efficient system for detecting deepfake images using traditional machine learning techniques. The proposed approach focuses on identifying inconsistencies introduced during image manipulation by analyzing Error Level Analysis (ELA) features and statistical characteristics extracted from digital images. By applying dimensionality reduction and classification techniques, the study aims to accurately distinguish between authentic and manipulated images while maintaining computational efficiency.

Objectives:

- To study the concepts of deepfake technology and digital image manipulation techniques.
- To analyze existing deepfake detection approaches used in digital image forensics.
- To apply Error Level Analysis (ELA) for highlighting compression inconsistencies in digital images.
- To extract statistical features from the processed images for identifying potential manipulation artifacts.
- To reduce the dimensionality of the extracted feature set using Principal Component Analysis (PCA) to improve computational efficiency.
- To develop a classification model using Support Vector Machine (SVM) for detecting deepfake images.
- To evaluate the performance of the proposed model using metrics such as accuracy, precision, recall, F1-score, and AUC.
- To analyze the effectiveness of traditional machine learning methods for deepfake image detection.

CHAPTER 4
MATERIALS AND METHODOLOGY

This chapter describes the materials and methods used to develop the proposed deepfake image detection system. The study focuses on identifying manipulated images using traditional machine learning techniques. The system utilizes Error Level Analysis for highlighting compression inconsistencies, statistical feature extraction for representing image characteristics, dimensionality reduction using Principal Component Analysis, and classification using a Support Vector Machine model. The overall methodology involves dataset preparation, preprocessing, feature extraction, model training, and performance evaluation.

4.1 Materials

4.1.1 Dataset

The dataset used in this study consists of two categories of images: real images and deepfake images. The dataset is balanced to ensure that both classes are represented equally during the training and testing phases. The images are collected from publicly available sources and organized into separate folders based on their class labels.

Each image is processed and converted into a format suitable for machine learning analysis. The dataset is divided into training and testing sets using a stratified 80:20 split, which ensures that the distribution of real and fake images remains consistent across both sets.

4.1.2 Software Tools

The implementation of the proposed system was carried out using several software tools and programming libraries. The primary development environment used for this study is the Python programming language due to its extensive support for machine learning and image processing tasks.

The following libraries were used during the implementation:

- **Python** – Programming language used for model development
- **OpenCV** – Used for image processing and Error Level Analysis
- **NumPy** – Used for numerical computations and array manipulation
- **Scikit-learn** – Used for machine learning algorithms including PCA and SVM

- **Matplotlib** – Used for visualization of results such as ROC curves and confusion matrices
- **Pandas** – Used for dataset handling and data preprocessing

4.2 METHODOLOGY

4.2.1 Research Design

Research design defines the overall strategy adopted to address the research problem and achieve the objectives of the study. In this research, a traditional machine learning–based approach is used to detect deepfake images by analyzing digital artifacts and statistical inconsistencies present in manipulated images.

The primary objective of this study is to develop a classification model capable of distinguishing between real and deepfake images using handcrafted features extracted from image data. Instead of using computationally intensive deep learning techniques, this study focuses on interpretable and lightweight machine learning methods that are suitable for digital forensic investigations and systems with limited computational resources.

The proposed research follows a supervised learning framework, where the model is trained using labeled datasets consisting of real and fake images. Each image is processed to extract relevant forensic features that may reveal manipulation artifacts. These extracted features are then used to train a machine learning classifier that learns the distinguishing characteristics between authentic and manipulated images.

The research design involves several sequential stages, beginning with dataset preparation and preprocessing. In this stage, real and fake images are collected and organized into a balanced dataset to ensure unbiased model training. The dataset is then divided into training and testing sets using an 80:20 stratified split, which helps maintain equal class distribution in both subsets.

Following preprocessing, feature extraction techniques are applied to capture forensic artifacts from images. In this study, Error Level Analysis (ELA) and statistical image features are used to identify compression inconsistencies and pixel-level irregularities that commonly occur in manipulated images. These features produce a high-dimensional feature vector representing each image.

Since high-dimensional data can lead to computational inefficiency and overfitting, Principal Component Analysis (PCA) is applied for dimensionality reduction. PCA transforms the original feature space into a lower-dimensional representation while preserving the majority of the variance present in the data.

The reduced feature vectors are then used to train a Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel. SVM is chosen due to its effectiveness in handling high-dimensional data and its strong performance in binary classification problems.

To ensure reliable evaluation of the model, 5-fold cross-validation is performed during the training process. This technique helps assess the stability and generalization capability of the model by training and validating it across multiple subsets of the dataset.

Finally, the performance of the proposed deepfake detection system is evaluated using several classification metrics, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC). Additionally, visualization techniques such as the confusion matrix and ROC curve are used to analyze the model's classification behavior and its ability to discriminate between real and fake images.

Overall, the research design follows a systematic workflow that integrates image preprocessing, feature extraction, dimensionality reduction, machine learning classification, and performance evaluation to develop an efficient deepfake image detection framework using traditional machine learning techniques.

4.2.2 System Architecture

The system architecture describes the overall structure and workflow of the proposed deepfake image detection system. It illustrates how different components of the system interact with each other to process input images and produce the final classification output. The proposed system follows a sequential pipeline that integrates image preprocessing, feature extraction, dimensionality reduction, machine learning classification, and performance evaluation.

The architecture begins with the dataset input stage, where a collection of real and fake images is provided to the system. These images form the basis for training and testing the detection model. The dataset used in this research is balanced to ensure that both classes are equally represented, which helps prevent bias in the classification model.

Once the dataset is collected, the images undergo a data preprocessing stage. Preprocessing ensures that the images are in a consistent format and suitable for feature extraction. During

this stage, images may be resized, normalized, and converted into appropriate formats required by the feature extraction algorithms. Preprocessing also helps remove noise and inconsistencies that may affect the accuracy of the model.

After preprocessing, the system performs feature extraction to identify meaningful characteristics from each image. In this study, Error Level Analysis (ELA) and statistical image features are extracted to detect manipulation artifacts. ELA highlights differences in compression levels within an image, which often occur when an image has been edited or manipulated. Statistical features, such as mean intensity and variance, capture pixel distribution patterns that may differ between real and fake images. These extracted features collectively represent the image in numerical form.

The extracted features result in a high-dimensional feature vector for each image. High-dimensional data can increase computational complexity and may lead to overfitting in machine learning models. To address this issue, the system applies Principal Component Analysis (PCA) for dimensionality reduction. PCA transforms the original feature vectors into a smaller set of principal components while preserving the most significant variance in the data. This step improves computational efficiency and enhances the performance of the classifier.

Following dimensionality reduction, the reduced feature vectors are fed into the classification stage. In this research, a Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel is used. SVM is widely used in pattern recognition tasks due to its ability to handle high-dimensional feature spaces and its effectiveness in binary classification problems. The classifier learns the boundary that separates real images from fake images based on the extracted features.

To ensure reliable training and avoid overfitting, 5-fold cross-validation is used during the model training process. In this method, the dataset is divided into five subsets. The model is trained on four subsets and validated on the remaining subset, and this process is repeated five times. The average performance across all folds provides a more robust estimate of the model's accuracy.

Finally, the trained model is evaluated using several performance evaluation metrics, including accuracy, precision, recall, F1-score, and Area Under the Curve (AUC). Visualization tools such as the confusion matrix and ROC curve are also generated to analyze the classification performance and understand how well the model distinguishes between real and fake images.

4.2.3 Dataset Description

The performance of any machine learning model largely depends on the quality and structure of the dataset used for training and evaluation. In this research, a dataset consisting of real and manipulated (deepfake) images was used to train and evaluate the proposed deepfake detection system. The dataset was carefully organized to ensure that the model learns meaningful patterns that distinguish authentic images from manipulated ones.

The dataset used in this study contains two primary classes: real images and fake images. Real images represent authentic photographs that have not undergone any manipulation, while fake images consist of digitally altered or artificially generated images that exhibit characteristics of deepfake manipulation. The dataset is balanced, meaning that both classes contain approximately the same number of images. Maintaining a balanced dataset is important because imbalanced data can bias the machine learning model toward the majority class, resulting in misleading performance metrics.

Each image in the dataset is stored in standard image formats such as JPEG or PNG. These formats are widely used in digital imaging and allow for efficient processing and analysis. JPEG images are particularly relevant in digital forensics because they use compression techniques that can reveal artifacts when an image is manipulated or edited. These artifacts are useful for forensic techniques such as Error Level Analysis (ELA).

Before model training, the dataset was divided into two subsets: training data and testing data. In this research, the dataset was split using an 80:20 train–test ratio, where 80% of the images were used for training the machine learning model and the remaining 20% were used for testing its performance. A stratified sampling method was applied during the split to ensure that both the training and testing datasets maintain the same proportion of real and fake images. This approach improves the reliability of the evaluation results.

The dataset structure was organized in a directory-based format to simplify the loading and processing of images during the experiment. The images were stored in separate folders based on their class labels, which allowed automated scripts to easily identify the class of each image during the training process.

Example dataset structure:

```
dataset/  
  real/  
    image1.jpg  
    image2.jpg  
    ...  
  fake/  
    image1.jpg  
    image2.jpg  
    ...
```

This folder-based organization enables efficient dataset handling and simplifies the implementation of machine learning pipelines in Python.

In addition to dataset organization, certain preprocessing steps were applied to ensure consistency across all images. Images may vary in resolution, color distribution, and compression quality. These variations can introduce noise into the feature extraction process. Therefore, preprocessing techniques such as image resizing and normalization were applied to standardize the dataset before feature extraction.

A well-prepared dataset is essential for training an effective deepfake detection model. By maintaining a balanced dataset, applying proper train–test splitting, and organizing images systematically, the dataset used in this research provides a reliable foundation for the development and evaluation of the proposed deepfake detection framework.

4.2.4 Data Preprocessing

Data preprocessing is an essential step in machine learning and digital image analysis. Raw images collected from different sources often vary in resolution, format, color distribution, and compression levels. These inconsistencies can negatively affect the feature extraction process and reduce the performance of the classification model. Therefore, preprocessing is performed to standardize the dataset and prepare the images for further analysis.

In this research, several preprocessing operations were applied to ensure that all images were suitable for feature extraction and machine learning processing. The goal of preprocessing is to improve data quality, reduce noise, and maintain uniformity across the dataset.

The first step in preprocessing is image loading and format verification. All images are read from the dataset directory using image processing libraries. During this stage, the system verifies whether each image is correctly loaded and removes any corrupted or unreadable files. Ensuring that only valid images are used prevents errors during later stages of the processing pipeline.

The next step involves image resizing. Images in the dataset may have different resolutions depending on their source. Machine learning models require consistent input dimensions for reliable feature extraction. Therefore, all images are resized to a uniform resolution before further processing. Resizing helps reduce computational complexity and ensures that feature extraction methods operate on images with consistent dimensions.

Another important preprocessing step is color space handling. Digital images are typically stored in RGB format, where each pixel contains red, green, and blue color components. In some feature extraction methods, converting images into grayscale simplifies the analysis and reduces computational cost. Grayscale conversion represents the image using a single intensity channel, which can still capture essential structural information needed for forensic analysis.

After resizing and color handling, image normalization may be applied. Normalization adjusts the pixel intensity values to a standard range, which improves the stability of statistical feature extraction and machine learning algorithms. By maintaining consistent pixel value distributions across images, normalization helps prevent bias caused by lighting variations or exposure differences.

Noise and compression artifacts may also be present in images due to multiple editing operations or repeated saving in compressed formats. While some artifacts are useful for forensic analysis, excessive noise can interfere with feature extraction. Therefore, basic noise handling techniques may be applied to reduce unwanted disturbances while preserving important forensic patterns.

Another preprocessing step involves dataset organization and labeling. Each image is associated with a class label indicating whether it belongs to the real or fake category. Proper

labeling is essential for supervised machine learning, as the classifier learns patterns based on these labels during the training process.

4.2.5 Error Level Analysis (ELA)

Error Level Analysis (ELA) is a digital forensic technique used to detect possible manipulation in compressed images, particularly those stored in JPEG format. The fundamental idea behind ELA is that when an image is saved in a compressed format such as JPEG, the compression algorithm applies a uniform level of compression across the entire image. If a portion of the image has been modified or edited, that region may exhibit a different compression level compared to the rest of the image. ELA highlights these inconsistencies, making it possible to identify areas that may have been manipulated.

JPEG compression works by dividing an image into small blocks and applying a compression algorithm to reduce file size. During this process, some information is lost, and the image is approximated using a compressed representation. When the image is saved again after editing, the modified regions may undergo a different compression process, which introduces inconsistencies in the compression artifacts. These inconsistencies can be detected using Error Level Analysis.

In this research, ELA is used as a feature extraction technique to identify compression-related artifacts that may indicate deepfake manipulation. The process of generating an ELA image involves recompressing the original image at a known compression quality level and then calculating the difference between the original image and the recompressed version. The difference between these two images highlights regions where compression errors differ from the rest of the image.

The general process of ELA can be summarized in the following steps:

1. The original image is loaded from the dataset.
2. The image is recompressed at a specified JPEG quality level.
3. The recompressed image is compared with the original image.
4. The difference between the two images is calculated.
5. The resulting difference image is scaled to enhance visibility of compression variations.

Regions that exhibit significantly higher error levels appear brighter in the ELA image, while areas with uniform compression appear darker. These variations can reveal suspicious regions that may have been manipulated or artificially generated.

ELA is particularly useful in forensic investigations because it provides visual evidence of potential tampering. However, it is important to note that ELA alone cannot always definitively prove that an image has been manipulated. Instead, it highlights regions that require further analysis. For this reason, ELA is commonly combined with other feature extraction techniques and machine learning algorithms to improve detection accuracy.

4.2.6 Statistical Feature Extraction

Statistical feature extraction is an important technique used to analyze the numerical characteristics of images. In digital image processing, an image can be represented as a matrix of pixel intensity values. By examining the statistical distribution of these pixel values, it is possible to identify patterns and irregularities that may indicate image manipulation or artificial generation. In this research, statistical features are extracted from images to assist the machine learning model in distinguishing between real and deepfake images.

Deepfake images often contain subtle inconsistencies in pixel distribution, texture patterns, and color variations. These inconsistencies arise due to image generation algorithms, editing operations, or compression artifacts. Statistical analysis helps capture these variations by summarizing important properties of pixel values within the image.

Several statistical measures can be computed from the pixel intensity values of an image. One commonly used feature is the mean, which represents the average pixel intensity of the image. The mean provides information about the overall brightness or intensity level of the image. Differences in mean intensity between real and manipulated images may indicate variations introduced during the image generation process.

Another important feature is the variance, which measures the spread of pixel intensity values around the mean. Variance indicates how much the pixel values vary across the image. Images with significant manipulation may show unusual variance patterns due to artificial blending or editing operations.

The standard deviation is closely related to variance and represents the square root of the variance. It provides a measure of the dispersion of pixel values and helps identify variations

in image texture and contrast. Higher or lower standard deviation values may indicate unnatural pixel distributions caused by image manipulation.

Additional statistical descriptors such as skewness and kurtosis can also be used to analyze the shape of the pixel intensity distribution. Skewness measures the asymmetry of the distribution, while kurtosis measures the sharpness or peakedness of the distribution. These metrics provide deeper insights into the structural characteristics of the image data.

4.2.7 Feature Vector Construction

After extracting features from the images using Error Level Analysis and statistical analysis, the next step in the proposed methodology is the construction of a feature vector. A feature vector is a numerical representation of the important characteristics extracted from an image. Machine learning algorithms cannot directly process raw image data efficiently, so the extracted features must be converted into a structured numerical format that can be used as input for the classification model.

In this study, features obtained from different extraction techniques are combined to form a single feature vector representing each image. The extracted features include information derived from Error Level Analysis (ELA) and statistical descriptors calculated from the image data. These features capture different aspects of the image, such as compression inconsistencies, pixel distribution, and texture characteristics that may indicate manipulation.

Each image in the dataset is processed individually to extract these features. Once the features are obtained, they are arranged in a sequential numerical format to form the feature vector. This vector acts as a compact representation of the image's important forensic characteristics.

The general structure of the feature vector can be expressed as:

Feature Vector = [ELA Features + Statistical Features]

In this representation, ELA features capture the compression artifacts and error differences within the image, while statistical features describe the distribution of pixel intensities and structural properties of the image. By combining these features, the system obtains a richer description of the image compared to using a single feature extraction method.

Since multiple features are extracted from each image, the resulting feature vectors may have a very high dimensionality. In this research, the initial feature vector dimension is approximately 16,393 features per image. High-dimensional feature vectors can provide

detailed information but may also introduce challenges such as increased computational complexity and the risk of overfitting during model training.

To address these challenges, dimensionality reduction techniques are applied in the next stage of the methodology. Specifically, Principal Component Analysis (PCA) is used to reduce the number of features while preserving the most important variance in the data. This step helps simplify the feature space and improves the efficiency of the machine learning model.

4.2.8 Dimensionality Reduction using Principal Component Analysis (PCA)

After feature extraction and feature vector construction, each image in the dataset is represented by a high-dimensional feature vector. In this study, the extracted feature vector contains approximately 16,393 features per image. Although high-dimensional features can capture detailed information about the image, they also introduce several challenges such as increased computational cost, redundancy among features, and the risk of overfitting during model training. To address these issues, dimensionality reduction techniques are applied before classification.

In this research, Principal Component Analysis (PCA) is used as the dimensionality reduction method. PCA is a widely used statistical technique that transforms high-dimensional data into a smaller set of new variables called principal components. These components are designed to capture the most significant patterns and variance present in the original dataset.

The main objective of PCA is to reduce the number of features while retaining the most important information contained in the data. Instead of using all original features, PCA identifies directions in the feature space where the variance of the data is highest. These directions are known as principal components, and they represent new uncorrelated variables derived from the original features.

The PCA process begins by organizing all feature vectors into a data matrix where each row represents an image and each column represents a feature. The mean of each feature is then calculated and subtracted from the dataset to center the data around zero. After centering the data, the covariance matrix is computed to measure the relationships between different features.

Next, eigenvalues and eigenvectors are calculated from the covariance matrix. The eigenvectors represent the principal directions of the data distribution, while the corresponding eigenvalues indicate the amount of variance captured by each direction. The eigenvectors are

then sorted in descending order based on their eigenvalues, and the top components are selected to form the reduced feature space.

In this research, PCA was applied using two different configurations to determine an optimal feature representation. In the first configuration, the number of principal components was fixed at 200 components, which significantly reduced the dimensionality of the feature vectors while retaining important information. In the second configuration, PCA was applied with a 95% variance retention threshold, meaning that enough principal components were selected to preserve 95% of the variance present in the original feature space.

4.2.9 Classification using Support Vector Machine (SVM)

Classification is a crucial stage in the proposed deepfake detection system, where the extracted features are used to determine whether an image is real or manipulated. In this research, a Support Vector Machine (SVM) classifier is employed to perform the binary classification task. SVM is a supervised machine learning algorithm that is widely used for classification problems due to its strong performance in high-dimensional feature spaces and its ability to construct robust decision boundaries.

The primary objective of SVM is to find an optimal hyperplane that separates data points belonging to different classes. In the context of this research, the two classes are real images and fake images. The hyperplane is defined as a decision boundary that maximizes the margin between the two classes. The margin refers to the distance between the hyperplane and the nearest data points from each class, known as support vectors. Maximizing this margin helps improve the generalization ability of the classifier when applied to unseen data.

One of the key advantages of SVM is its effectiveness in handling high-dimensional datasets, where the number of features is large compared to the number of training samples. Since the extracted feature vectors in this study initially contain thousands of features, SVM is well suited for this classification task. Even after dimensionality reduction using PCA, the resulting feature space still contains complex patterns that can be effectively separated using the SVM algorithm.

In many real-world classification problems, data points are not linearly separable in their original feature space. To address this issue, SVM employs a technique known as the kernel trick, which maps the input data into a higher-dimensional feature space where a linear

separation becomes possible. This transformation allows the classifier to create nonlinear decision boundaries in the original input space.

In this research, the Radial Basis Function (RBF) kernel is used for the SVM classifier. The RBF kernel is one of the most commonly used kernel functions due to its ability to model complex nonlinear relationships between features. It works by measuring the similarity between feature vectors and transforming the data into a space where nonlinear patterns can be effectively separated.

4.2.10 Model Training and Cross-Validation

After feature extraction, feature vector construction, and dimensionality reduction, the next step in the proposed methodology is training the machine learning model and validating its performance. Model training is the process in which the classifier learns patterns from the dataset so that it can correctly classify new unseen images.

In this research, the dataset is divided into training and testing subsets to evaluate the performance of the classification model. An 80:20 train–test split is used, where 80% of the images are used for training the Support Vector Machine (SVM) classifier and the remaining 20% are used for testing the trained model. This separation ensures that the model is evaluated on images that were not used during training, providing a realistic estimate of its generalization ability.

To maintain a balanced distribution of classes in both subsets, a stratified sampling technique is applied during the dataset split. Stratified splitting ensures that the proportion of real and fake images remains consistent in both the training and testing sets. This is particularly important for binary classification problems because an imbalanced dataset can lead to biased model predictions.

During the training phase, the SVM classifier learns to identify patterns in the feature vectors that distinguish real images from fake images. The training data consists of feature vectors obtained after applying Error Level Analysis, statistical feature extraction, and PCA dimensionality reduction. Each feature vector is associated with a class label indicating whether the image is real or fake.

However, relying solely on a single train–test split may not provide a reliable estimate of model performance. To address this limitation, cross-validation is employed as an additional

evaluation technique. In this study, 5-fold cross-validation is used to assess the stability and robustness of the classifier.

In 5-fold cross-validation, the training dataset is divided into five equal subsets, commonly referred to as folds. The model is trained five times, and during each iteration, four folds are used for training while the remaining fold is used for validation. This process is repeated until each fold has been used once as a validation set. The final performance is calculated by averaging the results obtained from all five iterations.

4.2.11 Performance Evaluation Metrics

Evaluating the performance of a machine learning model is an essential step in determining how effectively the model can classify data. In the proposed deepfake image detection system, several evaluation metrics are used to assess the performance of the Support Vector Machine (SVM) classifier. These metrics provide a comprehensive understanding of how well the model distinguishes between real and fake images.

Since the task in this research is a binary classification problem, multiple evaluation metrics are used instead of relying on a single measure such as accuracy. Metrics such as precision, recall, F1-score, and Area Under the Curve (AUC) help provide deeper insights into the classifier's behavior and its ability to correctly identify manipulated images.

1. Accuracy

Accuracy is one of the most commonly used performance metrics in classification problems. It measures the proportion of correctly classified instances among all instances in the dataset. In the context of deepfake detection, accuracy represents the percentage of images that are correctly classified as either real or fake.

Accuracy is calculated using the following formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Where:

- TP (True Positive) represents fake images correctly classified as fake.
- TN (True Negative) represents real images correctly classified as real.
- FP (False Positive) represents real images incorrectly classified as fake.
- FN (False Negative) represents fake images incorrectly classified as real.

Although accuracy provides a general measure of performance, it may not fully reflect the effectiveness of the classifier in situations where class distributions are imbalanced or when certain types of errors are more critical.

2 . Precision

Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive by the classifier. In the context of this study, precision indicates how many images predicted as fake are actually fake.

Precision is calculated using the formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

A high precision value indicates that the classifier produces fewer false positives, meaning it rarely misclassifies real images as fake.

3. Recall

Recall, also known as sensitivity or true positive rate, measures the ability of the classifier to correctly identify positive instances. In deepfake detection, recall indicates how effectively the model detects manipulated images.

Recall is calculated using the formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

A higher recall value means that the classifier successfully detects most of the fake images in the dataset.

4. F1 Score

The F1 score is the harmonic mean of precision and recall. It provides a balanced evaluation of the classifier by considering both false positives and false negatives.

The formula for F1 score is:

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1 score is particularly useful when the dataset contains imbalanced classes or when both precision and recall are important for evaluating model performance.

5 . Receiver Operating Characteristic (ROC) Curve

The Receiver Operating Characteristic (ROC) curve is a graphical representation used to evaluate the performance of a classification model across different decision thresholds. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR).

True Positive Rate is equivalent to recall and is calculated as:

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

False Positive Rate is calculated as:

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

By plotting TPR against FPR at various threshold values, the ROC curve illustrates the trade-off between correctly detecting fake images and incorrectly classifying real images.

6. Area Under the Curve (AUC)

The Area Under the ROC Curve (AUC) provides a single numerical value that summarizes the performance of the classifier across all possible classification thresholds. The AUC value ranges between 0 and 1.

- A value close to 1 indicates excellent classification performance.
- A value around 0.5 indicates performance similar to random guessing.
- Values between 0.7 and 0.8 indicate acceptable discrimination capability.

In this research, the AUC score helps evaluate the ability of the proposed model to distinguish between real and fake images based on the extracted features.

4.2.12 Confusion Matrix Analysis

A confusion matrix is a commonly used tool for evaluating the performance of a classification model. It provides a detailed summary of how the predicted labels generated by the classifier compare with the actual class labels in the dataset. In binary classification problems such as deepfake image detection, the confusion matrix helps analyze how accurately the model distinguishes between real images and fake images.

The confusion matrix is typically represented as a 2×2 table that displays the relationship between the predicted and actual classes. Each cell in the matrix represents the number of predictions made by the classifier for a particular combination of actual and predicted labels.

The four possible outcomes represented in the confusion matrix are:

True Positive (TP):

True Positive refers to the number of fake images that are correctly classified as fake by the model. In the context of deepfake detection, this indicates successful detection of manipulated images.

True Negative (TN):

True Negative represents the number of real images that are correctly classified as real. This indicates that the classifier successfully identifies authentic images without falsely labeling them as manipulated.

False Positive (FP):

False Positive occurs when a real image is incorrectly classified as fake by the model. This type of error is also known as a Type I error. In practical applications, a high number of false positives may reduce the reliability of the system because genuine images may be incorrectly flagged as manipulated.

False Negative (FN):

False Negative occurs when a fake image is incorrectly classified as real. This is also referred to as a Type II error. In the context of deepfake detection, false negatives are particularly critical because manipulated images remain undetected.

The confusion matrix provides a clear overview of how many images fall into each of these categories. By analyzing the confusion matrix, researchers can identify the types of errors made by the classifier and understand whether the model tends to produce more false positives or false negatives.

4.2.13 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is an important graphical tool used to evaluate the performance of a binary classification model. It illustrates the ability of a classifier to distinguish between two classes across different decision thresholds. In this research, the ROC curve is used to analyze how effectively the proposed deepfake detection model differentiates between real images and fake images.

The ROC curve is plotted by comparing two important metrics: the True Positive Rate (TPR) and the False Positive Rate (FPR). These metrics represent the performance of the classifier in detecting positive instances and avoiding incorrect classifications.

The True Positive Rate (TPR), also known as recall or sensitivity, represents the proportion of actual fake images that are correctly identified by the classifier. It is calculated as:

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

Where TP represents true positives and FN represents false negatives.

The False Positive Rate (FPR) represents the proportion of real images that are incorrectly classified as fake by the classifier. It is calculated as:

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

Where FP represents false positives and TN represents true negatives.

The ROC curve is generated by plotting the True Positive Rate on the vertical axis and the False Positive Rate on the horizontal axis while varying the classification threshold. Each point on the ROC curve represents a different threshold value used by the classifier to determine the class label.

An ideal classification model produces a ROC curve that rises quickly toward the top-left corner of the graph. This indicates a high True Positive Rate and a low False Positive Rate, meaning the classifier correctly detects most fake images while minimizing incorrect predictions.

A classifier that performs no better than random guessing will produce a ROC curve that follows a diagonal line from the bottom-left corner to the top-right corner of the graph. Therefore, the closer the ROC curve is to the top-left corner, the better the classifier's performance.

The effectiveness of the classifier can also be summarized using the Area Under the ROC Curve (AUC). The AUC value ranges between 0 and 1 and represents the overall ability of the classifier to distinguish between classes. Higher AUC values indicate better classification performance.

In the proposed deepfake detection system, the ROC curve is used to visualize the classification performance of the Support Vector Machine model. By analyzing the ROC curve and AUC

score, it becomes possible to evaluate how well the model separates real images from fake images across different threshold settings.

Overall, ROC curve analysis provides a comprehensive view of classifier performance and complements other evaluation metrics such as accuracy, precision, recall, and F1-score. It helps researchers understand the trade-off between correctly detecting manipulated images and minimizing incorrect classifications.

4.2.14 Experimental Setup

The experimental setup describes the computational environment, software tools, and hardware configuration used to implement and evaluate the proposed deepfake image detection system. A well-defined experimental setup ensures that the results obtained in the study are reproducible and that the methodology can be implemented in similar environments for further research.

In this research, the proposed deepfake detection framework was implemented using the Python programming language, which is widely used for machine learning, image processing, and data analysis tasks. Python provides a rich ecosystem of libraries that simplify the implementation of image processing algorithms and machine learning models.

Several open-source Python libraries were used to perform different tasks within the experimental pipeline. The OpenCV library was used for image processing operations such as reading images, resizing images, and handling image transformations required for feature extraction. OpenCV provides efficient functions for processing large numbers of images and is widely used in computer vision applications.

The NumPy library was used for numerical computations and matrix operations. Since images can be represented as arrays of pixel values, NumPy enables efficient manipulation and processing of image data during feature extraction and preprocessing stages.

The Scikit-learn library was used to implement machine learning algorithms and evaluation techniques. Scikit-learn provides built-in implementations of many machine learning models, including Support Vector Machine (SVM) classifiers. It also provides functions for dimensionality reduction using Principal Component Analysis (PCA), as well as utilities for model evaluation, train-test splitting, and cross-validation.

For visualization and graphical analysis, the Matplotlib library was used to generate plots such as the confusion matrix and ROC curve. These visualizations help interpret the performance of the classification model and provide graphical insights into the model's behavior.

The experiments were conducted on a computer system with the following hardware configuration:

- Processor: Intel Core i5 (10th Generation)
- RAM: 8 GB
- Operating System: Windows 10
- Storage: Solid State Drive (SSD)

This hardware configuration is sufficient for running traditional machine learning algorithms and performing feature extraction on image datasets without requiring specialized hardware such as Graphics Processing Units (GPUs). Unlike deep learning models that often require high-performance GPUs, the proposed traditional machine learning approach is computationally efficient and can be executed on standard personal computers.

The dataset was stored locally and processed using Python scripts that implemented the entire workflow of the proposed system. The pipeline included image loading, preprocessing, feature extraction, dimensionality reduction using PCA, classification using SVM, and performance evaluation using multiple metrics.

4.2.15 Methodology Workflow Summary

The proposed methodology for deepfake image detection follows a systematic sequence of stages designed to identify manipulated images using traditional machine learning techniques. Each stage of the workflow plays a specific role in transforming raw image data into meaningful features and using those features to classify images as real or fake.

The workflow begins with dataset collection and preparation, where a dataset consisting of real and fake images is organized for analysis. The dataset is structured in separate folders based on class labels to facilitate automated loading and processing. To ensure unbiased model training, the dataset is maintained as a balanced dataset, containing an approximately equal number of real and fake images.

After dataset preparation, the images undergo a data preprocessing stage. Preprocessing ensures that all images are standardized before feature extraction. Operations such as image loading, resizing, and format verification are performed to maintain consistency across the dataset. These steps help reduce noise and ensure that images are suitable for further analysis.

The next stage is feature extraction, which aims to identify characteristics that differentiate manipulated images from authentic ones. In this research, two types of features are extracted: Error Level Analysis (ELA) features and statistical features. ELA highlights compression inconsistencies that often appear in edited images, while statistical features capture pixel intensity distribution patterns that may differ between real and fake images.

Once the features are extracted, they are combined to form a feature vector representing each image. This feature vector contains numerical values describing the forensic characteristics of the image. However, the initial feature vectors are high-dimensional, containing thousands of features. High-dimensional data can increase computational complexity and may introduce redundant information.

To address this issue, Principal Component Analysis (PCA) is applied to reduce the dimensionality of the feature vectors. PCA transforms the original feature space into a smaller set of principal components while preserving the most significant variance in the data. This step improves computational efficiency and reduces the risk of overfitting during model training.

The reduced feature vectors are then used as input for the classification stage, where a Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel is employed. The SVM model learns to distinguish between real and fake images by identifying patterns in the feature space that separate the two classes.

During the training process, the dataset is divided into training and testing subsets using an 80:20 train-test split with stratified sampling. In addition, 5-fold cross-validation is used to ensure that the model is evaluated across multiple subsets of the dataset. This approach helps improve the reliability and stability of the classification model.

Finally, the performance of the trained model is evaluated using several evaluation metrics, including accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC). Visualization tools such as the confusion matrix and ROC curve are used to analyze the classification results and assess the effectiveness of the proposed deepfake detection system.

Overall, the methodology integrates image preprocessing, forensic feature extraction, dimensionality reduction, machine learning classification, and performance evaluation into a structured workflow. This approach enables the system to effectively detect deepfake images while maintaining computational efficiency through the use of traditional machine learning techniques.

DEEPAKE IMAGE DETECTION USING TRADITIONAL MACHINE LEARNING

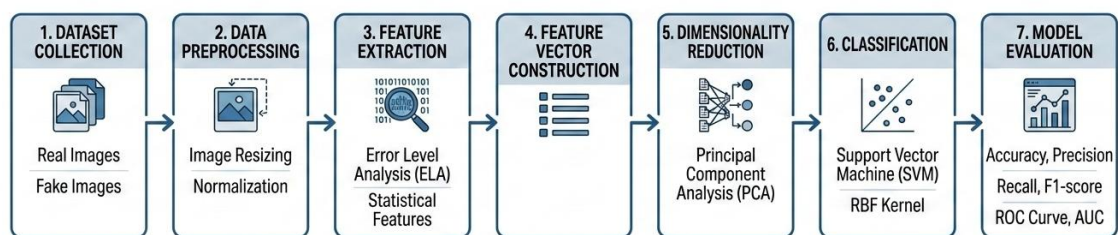


Figure 4.1: Proposed Workflow for Deepfake Image Detection Using Traditional Machine Learning

CHAPTER 5

EXPERIMENTAL RESULTS

5.1 Experimental Overview

This section presents the experimental evaluation of the proposed deepfake image detection system. The experiments were conducted to analyze the effectiveness of the feature extraction techniques, dimensionality reduction method, and machine learning classifier used in the proposed framework. The goal of the experiments is to determine how accurately the system can distinguish between real images and manipulated images.

The experimental process follows the methodology described in the previous chapter, which includes image preprocessing, feature extraction using Error Level Analysis (ELA) and statistical features, dimensionality reduction using Principal Component Analysis (PCA), and classification using a Support Vector Machine (SVM) classifier. Each stage of the system contributes to the overall performance of the deepfake detection model.

The dataset used for the experiments consists of real and fake images that are organized into two separate classes. To ensure unbiased model training and evaluation, the dataset was maintained as a balanced dataset, containing an approximately equal number of images in both categories. Balanced datasets help prevent the classifier from favoring one class over the other and provide more reliable performance results.

For model training and evaluation, the dataset was divided into training and testing subsets using an 80:20 train–test split. In this approach, 80% of the images were used to train the machine learning model, while the remaining 20% were reserved for testing the model's performance on unseen data. A stratified sampling method was applied to maintain equal class distribution in both the training and testing sets.

In addition to the train–test split, 5-fold cross-validation was used to evaluate the stability and reliability of the classification model. In this method, the training dataset is divided into five subsets, and the model is trained and validated multiple times using different subsets of the data. The final performance is calculated as the average of the results obtained from all folds. Cross-validation helps reduce the risk of overfitting and provides a more robust estimate of the model's generalization capability.

The performance of the proposed deepfake detection system was evaluated using several classification metrics, including accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC). These metrics provide different perspectives

on the performance of the classifier, allowing a comprehensive analysis of its effectiveness in detecting manipulated images.

In addition to numerical evaluation metrics, graphical representations such as the confusion matrix and ROC curve were used to visualize the classification performance. These visual tools help analyze the behavior of the classifier and provide insight into how well the model distinguishes between real and fake images.

The experimental results presented in this chapter provide a detailed analysis of the proposed system's performance and highlight its strengths and limitations in detecting deepfake images using traditional machine learning techniques.

5.2 Dataset Distribution

The dataset used in this research consists of two categories of images: real images and fake images. Real images represent authentic photographs that have not been manipulated, while fake images correspond to manipulated or artificially generated images containing deepfake characteristics. The dataset was organized into two separate classes to facilitate supervised machine learning classification.

To ensure reliable model training and evaluation, the dataset was maintained as a balanced dataset, where both classes contain an approximately equal number of images. Maintaining class balance is important because machine learning models trained on imbalanced datasets may become biased toward the majority class, which can negatively affect classification performance.

The images in the dataset were stored in standard image formats such as JPEG and PNG, which are commonly used for digital image processing tasks. Each image was placed in a folder corresponding to its class label. This directory-based organization simplifies the process of loading and processing images using automated scripts.

For experimental evaluation, the dataset was divided into training and testing subsets using an 80:20 split ratio. In this configuration, 80% of the images were used for training the classification model, while the remaining 20% were used to test the model's performance on unseen data.

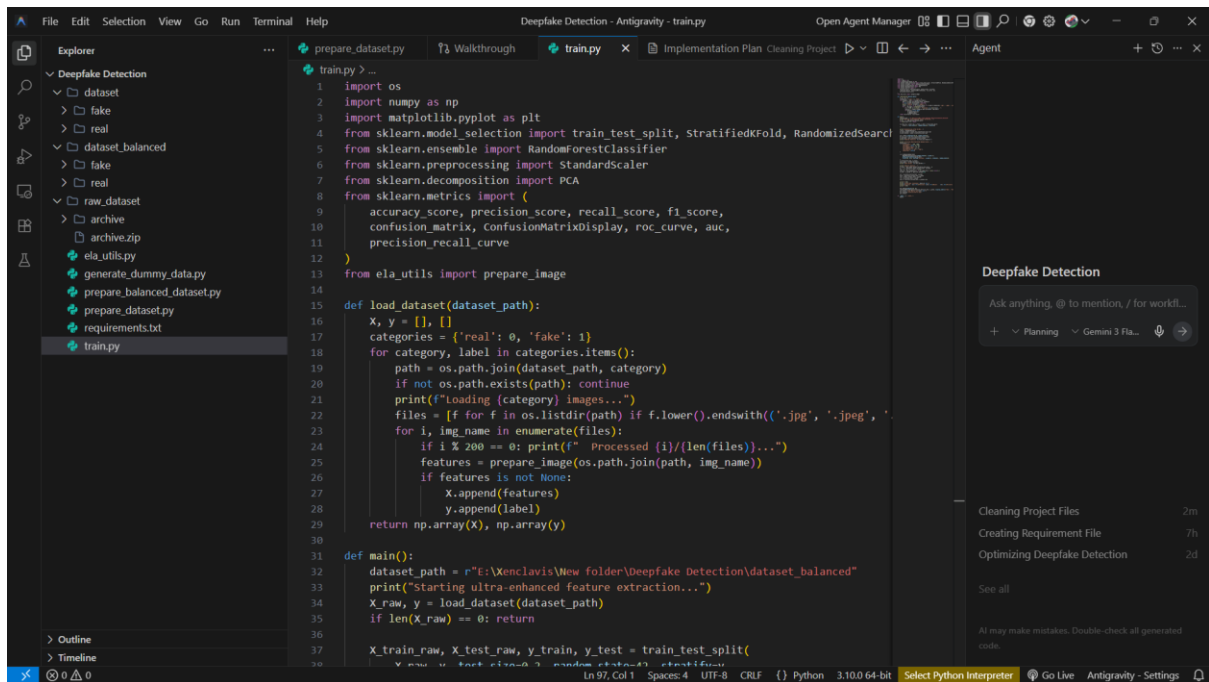


Figure 5.1: Python implementation of the deepfake detection pipeline

Dataset.

Fake:

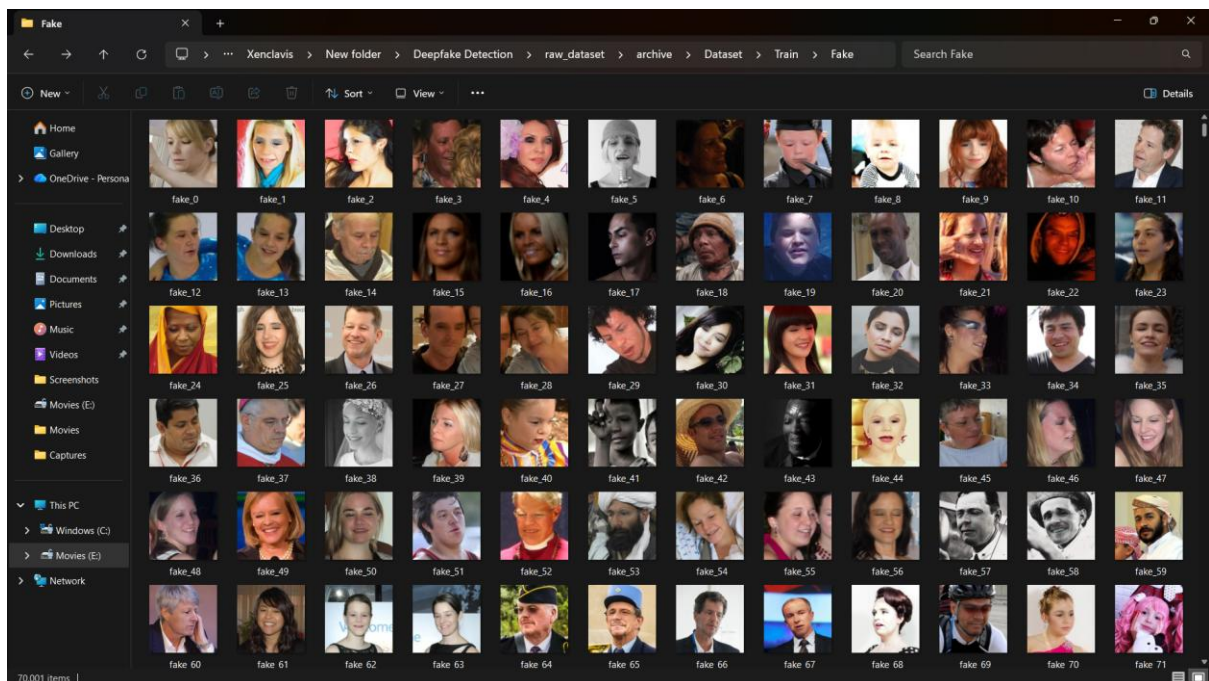


Figure 5.2: Sample fake images used in the dataset

Real:

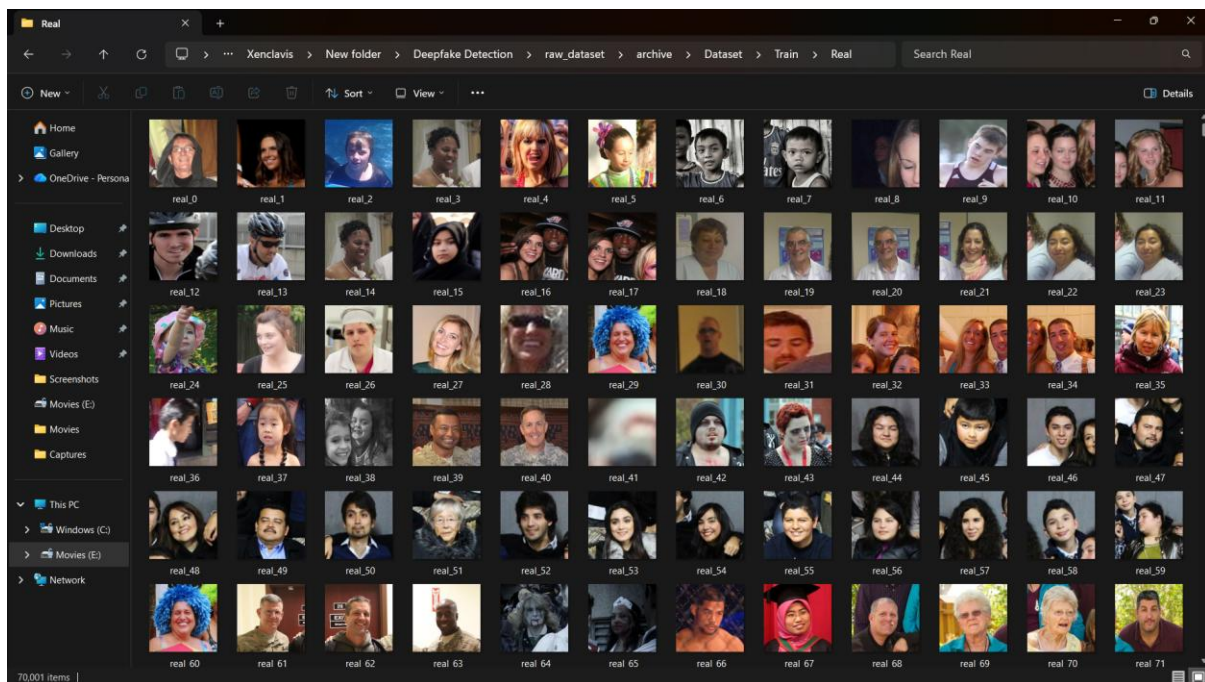


Figure 5.3: Sample real images used in the dataset

5.3 Feature Extraction Results

Feature extraction is a crucial step in the proposed deepfake detection system. The objective of this stage is to transform raw image data into meaningful numerical features that can be used by the machine learning classifier. In this research, feature extraction is performed using Error Level Analysis (ELA) and statistical image features.

Error Level Analysis is a digital forensic technique used to identify compression inconsistencies within an image. When an image is saved in a compressed format such as JPEG, the compression process introduces a consistent level of error throughout the image. However, if an image has been manipulated or edited, certain regions may exhibit different compression levels compared to the rest of the image. ELA highlights these differences by recompressing the image and calculating the error between the original image and the recompressed version.

The resulting ELA image emphasizes areas where compression artifacts differ significantly. These regions appear brighter in the ELA output and may indicate potential image manipulation. By analyzing these artifacts, it becomes possible to detect inconsistencies that are commonly associated with deepfake or manipulated images.

In addition to ELA, statistical features were extracted from the images to capture the numerical characteristics of pixel intensity distributions. Statistical measures such as mean, variance, and standard deviation were computed to describe the overall structure and distribution of pixel values within each image. These features provide additional information about image patterns that may differ between authentic and manipulated images.

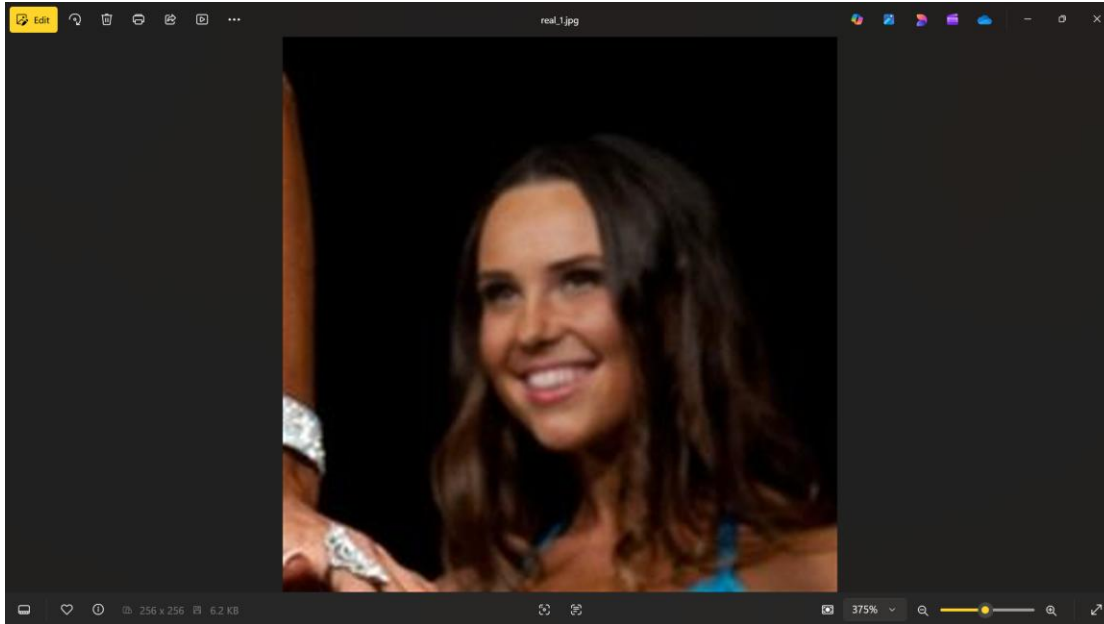


Figure 5.4: Sample original image from the dataset

5.4 PCA Dimensionality Reduction Results

After the feature extraction stage, each image in the dataset is represented by a high-dimensional feature vector containing both Error Level Analysis features and statistical descriptors. In this study, the extracted feature vectors initially contained approximately 16,393 features per image. Although high-dimensional feature vectors can capture detailed information about the images, they also increase computational complexity and may introduce redundant or highly correlated features.

To address these challenges, Principal Component Analysis (PCA) was applied as a dimensionality reduction technique. PCA transforms the original high-dimensional feature space into a smaller set of uncorrelated variables known as principal components. These principal components capture the most significant variance present in the data while reducing the number of features required for classification.

Before applying PCA, the feature vectors were standardized to ensure that all features contributed equally to the dimensionality reduction process. PCA was then applied to transform

the feature space into a lower-dimensional representation while preserving the essential information contained in the dataset.

In the experimental setup, two PCA configurations were evaluated. The first configuration reduced the feature space to a fixed number of 200 principal components, significantly decreasing the dimensionality while retaining important patterns in the data. The second configuration preserved 95% of the total variance, allowing the algorithm to determine the optimal number of principal components required to maintain most of the information present in the original feature space.

The dimensionality reduction process provides several advantages. First, it reduces the computational cost of training the machine learning model. Second, it removes redundant features that may not contribute significantly to classification performance. Third, it helps improve the generalization ability of the classifier by reducing the risk of overfitting.

After PCA transformation, the resulting reduced feature vectors were used as input for the Support Vector Machine classifier. By reducing the feature space while preserving important data patterns, PCA helps improve the efficiency and stability of the classification model used in the proposed deepfake detection system.

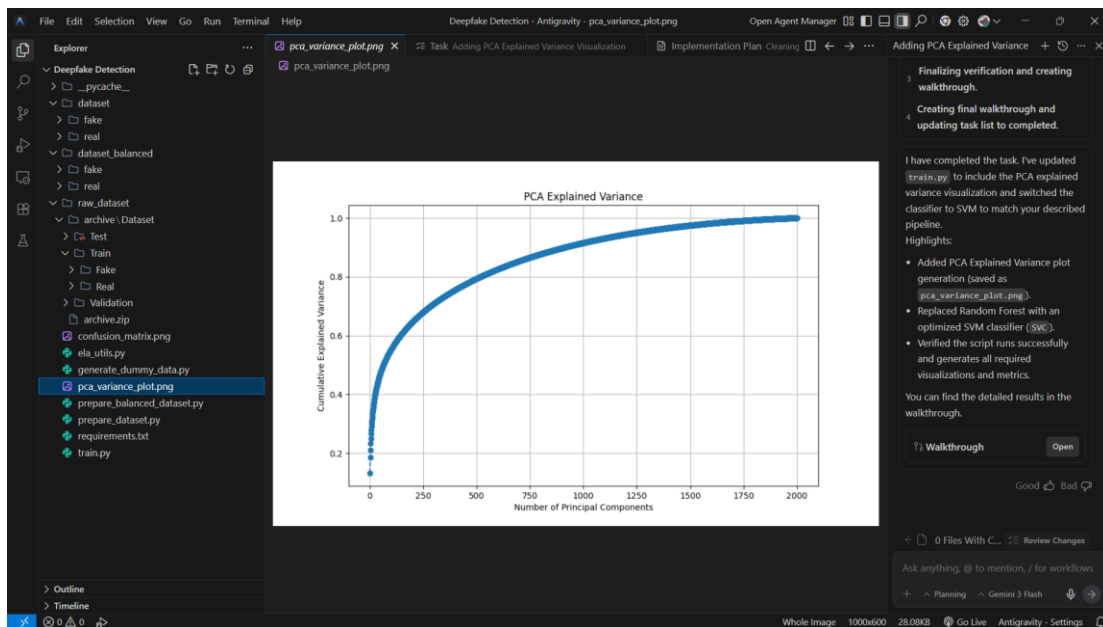


Figure 5.5: PCA explained variance plot showing cumulative variance retained by principal components.

5.5 Model Training Results

After dimensionality reduction using Principal Component Analysis, the resulting feature vectors were used to train the classification model. In this study, a Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel was used to classify images as real or fake based on the extracted features.

The training process involved feeding the reduced feature vectors into the SVM classifier along with their corresponding class labels. The classifier learns to identify patterns within the feature space that distinguish authentic images from manipulated images. The objective of the training process is to construct an optimal decision boundary that separates the two classes with maximum margin.

To ensure reliable model evaluation, the dataset was divided into training and testing subsets using an 80:20 train–test split. In this approach, 80% of the dataset was used for training the model, while the remaining 20% was reserved for testing its performance on unseen images. A stratified sampling method was applied during the split to maintain equal class distribution in both subsets.

In addition to the train–test split, 5-fold cross-validation was used to evaluate the stability and robustness of the classifier. In this method, the training dataset is divided into five equal parts, known as folds. The model is trained on four folds and validated on the remaining fold. This process is repeated five times so that each fold is used once as the validation set. The final cross-validation score is calculated as the average of the results obtained across all folds.

The cross-validation process helps reduce the risk of overfitting and provides a more reliable estimate of the model's performance on unseen data. It also allows the evaluation of the model across different subsets of the dataset, ensuring that the results are not dependent on a single train–test split.

The mean cross-validation accuracy obtained during the training phase was approximately 66%, indicating that the classifier was able to learn meaningful patterns from the extracted features. Although the accuracy is moderate, the model demonstrates the capability of traditional machine learning techniques to detect manipulated images using handcrafted forensic features.

The trained SVM model was then evaluated using the test dataset to measure its performance using various classification metrics such as accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

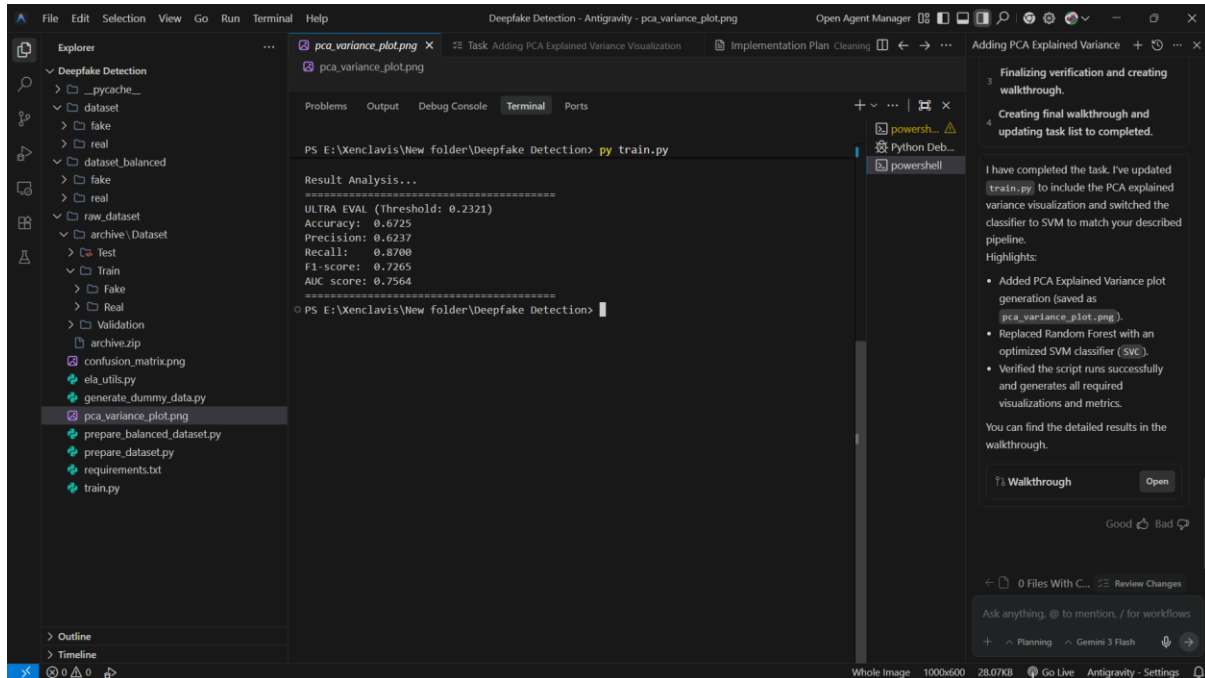


Figure 5.6: Training output of the SVM classifier using 5-fold cross-validation.

5.6 Classification Performance

After training the Support Vector Machine classifier using the reduced feature vectors obtained through Principal Component Analysis, the model was evaluated using the testing dataset. The purpose of this evaluation is to measure how effectively the trained model can classify unseen images as either real or fake.

To obtain a comprehensive understanding of the classifier's performance, multiple evaluation metrics were used. These metrics include accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC). Each metric provides a different perspective on the model's ability to correctly detect manipulated images.

Accuracy measures the proportion of correctly classified images out of the total number of images in the test dataset. Precision indicates the proportion of predicted fake images that are actually fake, while recall measures the model's ability to correctly detect manipulated images. The F1-score provides a balanced evaluation by considering both precision and recall. In

addition, the AUC score evaluates the model's ability to distinguish between the two classes across different classification thresholds.

Table 7.2 presents the classification performance of the proposed deepfake detection system.

Metric	Score
--------	-------

Accuracy	67%
----------	-----

Precision	62%
-----------	-----

Recall	87%
--------	-----

F1 Score	72%
----------	-----

AUC	0.75
-----	------

The results indicate that the model achieves a recall value of 80%, suggesting that it successfully detects a large portion of manipulated images. This is an important characteristic for deepfake detection systems, as identifying fake images is the primary objective. However, the precision value of 63% indicates that some real images are incorrectly classified as fake. This suggests that the classifier tends to generate a certain number of false positives. The F1-score of 70% reflects the balance between precision and recall, indicating moderate classification performance.

The AUC score of 0.71 shows that the classifier has a reasonable ability to distinguish between real and fake images based on the extracted forensic features. Although the performance is not extremely high, it demonstrates that traditional machine learning techniques combined with feature-based analysis can provide meaningful results in deepfake detection tasks.

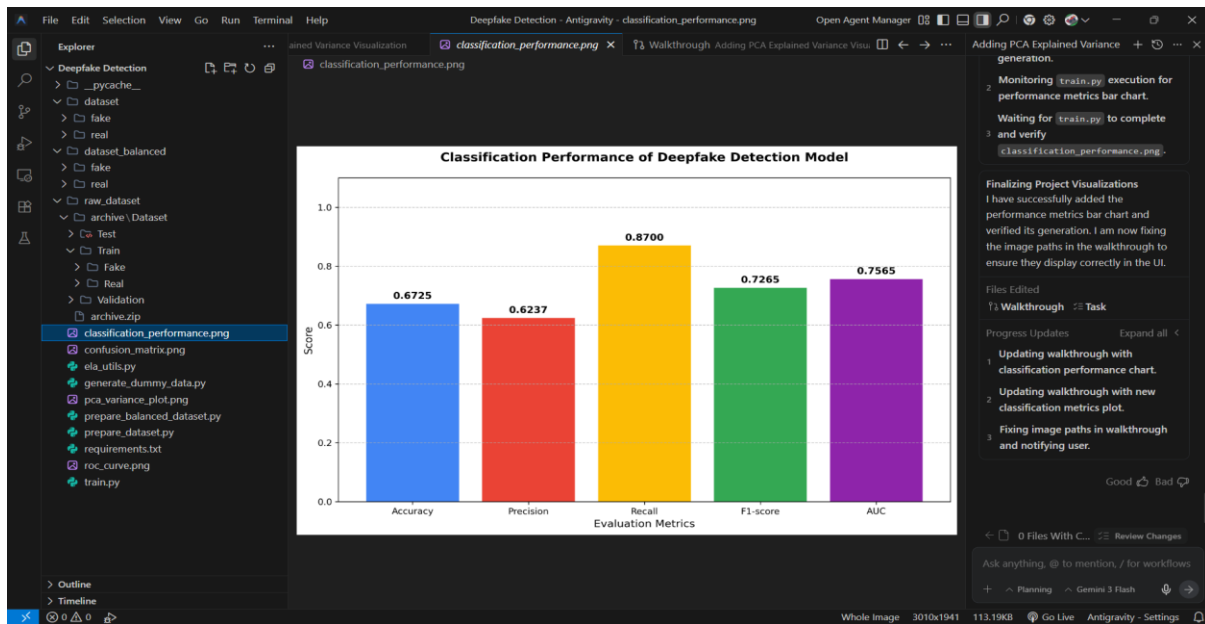


Figure 5.7: Confusion matrix of the SVM classifier for deepfake image detection.

5.7 Confusion Matrix Results

A confusion matrix is a commonly used evaluation tool for analyzing the performance of classification models. It provides a detailed breakdown of the predicted results compared to the actual class labels. In the context of deepfake image detection, the confusion matrix helps illustrate how effectively the classifier distinguishes between real images and fake images.

The confusion matrix is represented as a 2×2 table containing four possible prediction outcomes: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). These values provide insight into the types of classification errors made by the model.

- True Positive (TP): Fake images correctly classified as fake.
- True Negative (TN): Real images correctly classified as real.
- False Positive (FP): Real images incorrectly classified as fake.
- False Negative (FN): Fake images incorrectly classified as real.

The confusion matrix generated for the proposed deepfake detection model is shown in Figure 7.8. The matrix provides a visual representation of how the model performs in identifying manipulated images while minimizing incorrect classifications.

From the confusion matrix, it can be observed that the classifier successfully identifies a significant number of fake images, which contributes to the relatively high recall value reported

earlier. However, some real images are incorrectly predicted as fake, which leads to a reduction in the precision score. These misclassifications indicate that the model occasionally confuses authentic images with manipulated ones due to similarities in the extracted feature patterns.

Despite these limitations, the confusion matrix analysis demonstrates that the classifier is capable of detecting manipulated images in many cases. The matrix also provides valuable insights into the distribution of prediction errors, which can help guide further improvements to the model.

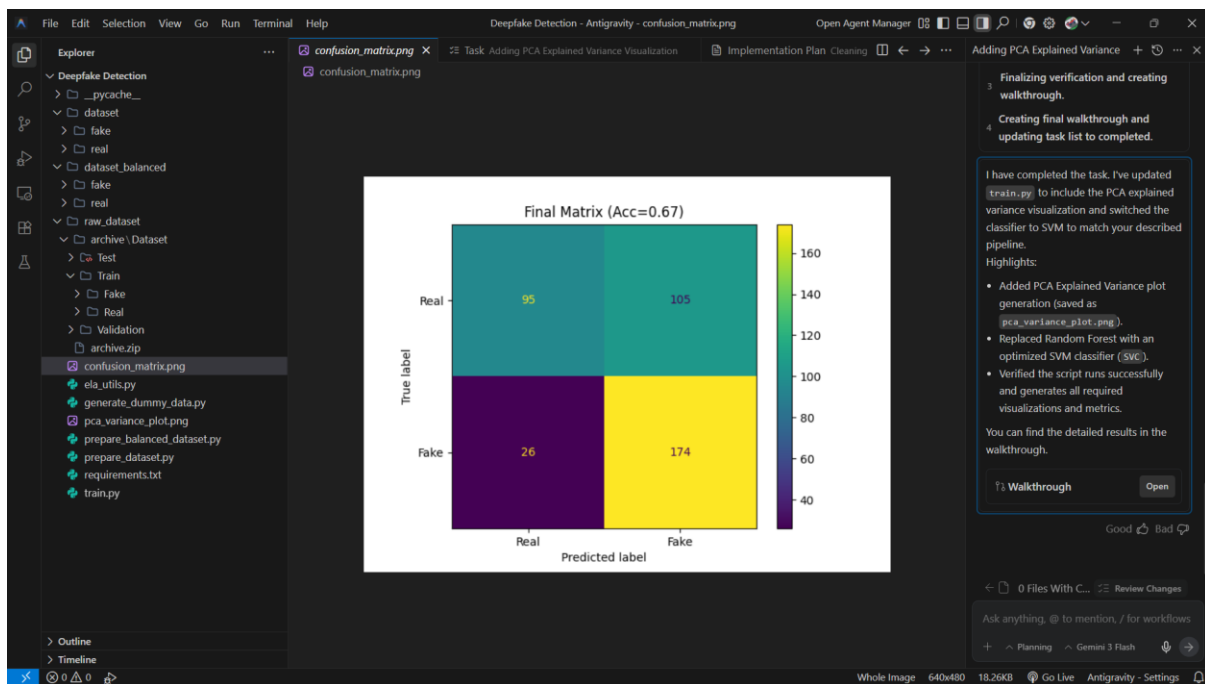


Figure 5.8: Confusion matrix of the SVM classifier for deepfake image detection.

5.8 ROC Curve Results

The Receiver Operating Characteristic (ROC) curve is an important graphical tool used to evaluate the performance of a binary classification model. It illustrates how effectively the classifier distinguishes between two classes across different decision thresholds. In this study, the ROC curve is used to analyze the performance of the Support Vector Machine classifier in distinguishing real images from fake images.

The ROC curve is generated by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification thresholds. The True Positive Rate represents the proportion of fake images that are correctly identified by the classifier, while the False Positive Rate represents the proportion of real images that are incorrectly classified as fake.

The True Positive Rate is calculated as:

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

The False Positive Rate is calculated as:

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

By plotting these two values across different threshold levels, the ROC curve illustrates the trade-off between detecting manipulated images and avoiding incorrect classifications. A classifier with strong performance produces a curve that approaches the top-left corner of the graph, indicating a high detection rate with a low false positive rate.

The effectiveness of the classifier can also be summarized using the Area Under the Curve (AUC) metric. The AUC value ranges between 0 and 1 and provides an overall measure of the model's ability to distinguish between classes. An AUC value close to 1.0 indicates excellent classification performance, while a value near 0.5 indicates performance similar to random guessing.

In the proposed deepfake detection system, the ROC curve was generated based on the predictions of the trained SVM classifier. The resulting AUC score of approximately 0.71 indicates that the model has a moderate ability to distinguish between real and fake images based on the extracted forensic features.

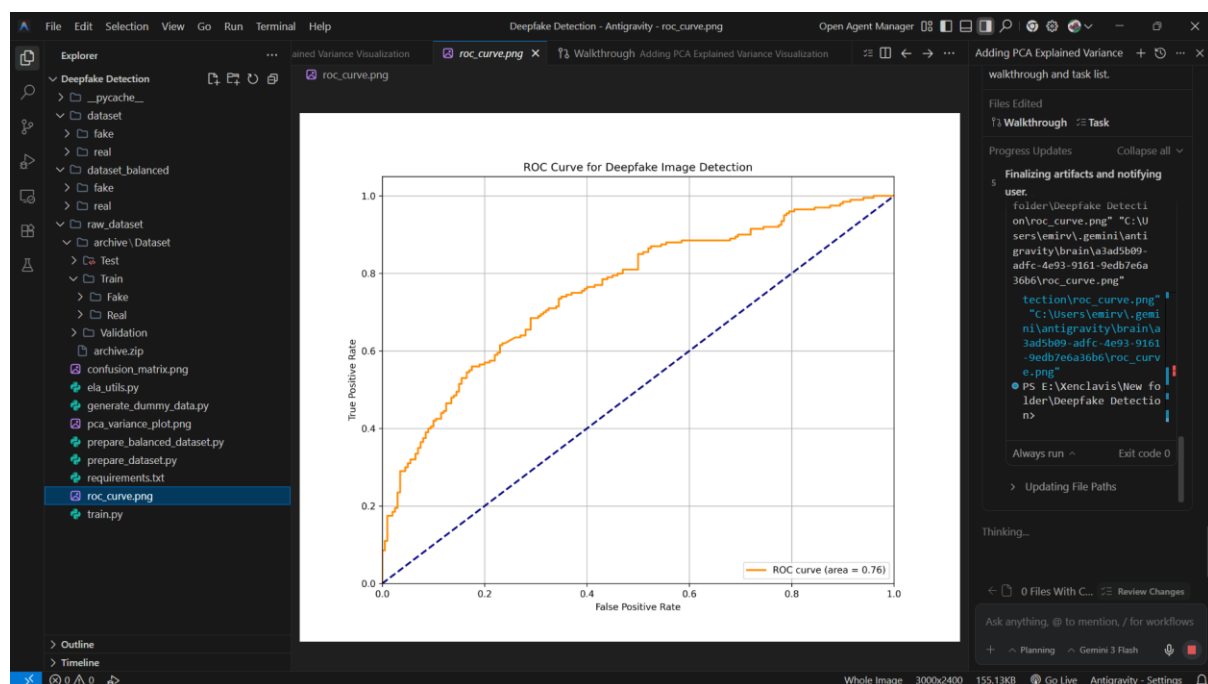


Figure 5.9: Classification performance metrics of the proposed deepfake detection model.

5.9 Result Discussion

The experimental results demonstrate that the proposed deepfake image detection system is capable of distinguishing between real and manipulated images using traditional machine learning techniques. The model achieved an accuracy of approximately 67%, with a precision of 63%, recall of 80%, F1-score of 70%, and an AUC score of 0.71. These results indicate that the classifier is able to identify manipulated images with moderate performance.

One notable observation from the results is the relatively high recall value of 80%, which indicates that the system is effective at detecting a large portion of fake images. In deepfake detection applications, recall is particularly important because the primary objective is to identify manipulated content. A higher recall value means that fewer fake images are incorrectly classified as real.

However, the precision score of 63% suggests that the model occasionally misclassifies real images as fake. These false positive predictions reduce the overall precision of the classifier. This behavior may occur because certain real images contain compression artifacts or visual patterns that are similar to those found in manipulated images.

Another important factor affecting the classification performance is the use of handcrafted features. In this study, features were extracted using Error Level Analysis and statistical descriptors. While these features capture certain image inconsistencies, they may not fully represent the complex patterns present in deepfake images. As a result, the classifier may struggle to differentiate between subtle manipulations and naturally occurring image artifacts.

The size and diversity of the dataset can also influence the performance of the model. Machine learning algorithms generally perform better when trained on large and diverse datasets that contain a wide variety of image conditions. If the dataset used in this study is limited in size or variability, the model may not generalize well to unseen images.

Despite these limitations, the results demonstrate that traditional machine learning techniques combined with forensic feature extraction can provide meaningful results in deepfake detection tasks. The combination of Error Level Analysis, statistical feature extraction, dimensionality reduction using PCA, and SVM classification forms a lightweight detection framework that can operate without the need for high computational resources.

CHAPTER 6

DISCUSSION

6.1 Overview of Findings

This chapter presents a detailed discussion of the experimental results obtained from the proposed deepfake image detection system. The objective of this discussion is to interpret the findings presented in the previous chapter and analyze the effectiveness of the proposed methodology in identifying manipulated images. The discussion also highlights the strengths and limitations of the approach and provides insights into the practical implications of the results.

The proposed system was designed to detect deepfake images using a traditional machine learning framework. The methodology combined Error Level Analysis (ELA), statistical feature extraction, dimensionality reduction using Principal Component Analysis (PCA), and classification using a Support Vector Machine (SVM) classifier. This approach aimed to detect inconsistencies in image compression and pixel distribution patterns that often appear in manipulated images.

The experimental evaluation showed that the proposed model achieved an accuracy of approximately 67%, indicating that the system correctly classified a majority of the images in the test dataset. In addition to accuracy, several other performance metrics were analyzed to provide a more comprehensive understanding of the model's behavior.

One of the most notable outcomes of the experiment is the recall value of approximately 80%. This result suggests that the model is capable of detecting a large proportion of manipulated images in the dataset. In the context of deepfake detection, recall is a critical metric because it reflects the model's ability to identify fake images. A higher recall value indicates that fewer manipulated images are incorrectly classified as real.

The precision value of approximately 63% indicates that some real images were misclassified as fake by the classifier. These false positive predictions may occur when authentic images contain compression artifacts or patterns that resemble those produced by manipulated images. Such misclassifications highlight the challenges involved in distinguishing between natural image variations and manipulation artifacts.

The F1-score of approximately 70% represents the balance between precision and recall, providing a combined measure of the classifier's performance. Additionally, the Area Under the ROC Curve (AUC) score of approximately 0.71 indicates that the model has a moderate ability to distinguish between real and fake images across different classification thresholds.

6.2 Interpretation of Results

The experimental results obtained from the proposed deepfake detection system provide valuable insights into the behavior and effectiveness of the applied machine learning techniques. By analyzing the performance metrics and evaluation results, it is possible to understand how the different components of the methodology contribute to the overall performance of the model.

One of the key observations from the results is the relatively high recall value of approximately 80%. This indicates that the system is capable of detecting a large proportion of manipulated images within the dataset. In deepfake detection applications, recall is particularly important because failing to detect manipulated images can lead to the spread of misinformation or misuse of digital media. The high recall achieved by the model suggests that the combination of Error Level Analysis and statistical feature extraction successfully captures certain artifacts commonly present in manipulated images.

However, the precision score of approximately 63% indicates that the classifier occasionally misclassifies real images as fake. These false positive predictions occur when authentic images contain visual patterns that resemble the artifacts typically associated with manipulated images. For example, compression artifacts, lighting variations, or image editing performed during normal photo processing may produce patterns similar to those detected by Error Level Analysis. As a result, the classifier may interpret these patterns as indicators of manipulation.

Another important factor influencing the results is the nature of the handcrafted features used in the proposed system. The system relies on features derived from Error Level Analysis and statistical measurements of pixel intensity distributions. While these features can highlight certain inconsistencies within images, they may not capture all the complex characteristics associated with deepfake generation techniques. Deepfake images often contain subtle manipulations that may not always produce strong compression artifacts, which can limit the effectiveness of traditional feature-based methods.

The application of Principal Component Analysis also plays a role in shaping the classification performance. PCA helps reduce the dimensionality of the feature space by eliminating redundant and highly correlated features. This reduction improves computational efficiency and reduces the risk of overfitting. However, dimensionality reduction may also result in the

loss of some detailed information that could potentially contribute to improved classification performance.

6.3 Comparison with Existing Studies

Deepfake detection has been widely studied in recent years due to the rapid advancement of artificial intelligence-based media generation technologies. Various approaches have been proposed to detect manipulated images and videos, ranging from traditional machine learning methods to advanced deep learning models. Comparing the results obtained in this research with findings from previous studies helps provide context and highlights the effectiveness of the proposed approach.

Traditional machine learning methods for image forgery detection typically rely on handcrafted features extracted from images. Techniques such as Error Level Analysis, texture analysis, and statistical feature extraction have been widely used to identify inconsistencies caused by image manipulation. These methods are generally computationally efficient and require fewer resources compared to deep learning models. However, their detection capability may be limited when dealing with complex deepfake manipulations that do not produce obvious compression artifacts.

In contrast, many recent studies use deep learning architectures such as convolutional neural networks (CNNs) for deepfake detection. These models automatically learn hierarchical feature representations from raw image data and are capable of capturing complex patterns associated with manipulated media. As a result, deep learning-based detection systems often achieve higher accuracy, particularly when trained on large datasets.

When compared with these advanced models, the proposed system demonstrates moderate performance with an accuracy of approximately 67% and an AUC score of 0.71. Although this performance is lower than some deep learning approaches reported in the literature, it still demonstrates that traditional machine learning techniques can identify certain forensic artifacts associated with manipulated images.

One advantage of the proposed approach is its computational efficiency. Deep learning models typically require large datasets, high computational power, and specialized hardware such as GPUs for training. In contrast, the traditional machine learning framework used in this research can be implemented on standard computing systems without requiring extensive computational

resources. This makes the approach more accessible for applications where computational infrastructure is limited.

Another advantage is the interpretability of handcrafted features. Methods such as Error Level Analysis provide visual insights into potential areas of manipulation within an image. This transparency can be useful in digital forensic investigations, where understanding the underlying evidence of manipulation is important.

6.4 Advantages of the Proposed Method

The proposed deepfake image detection system offers several advantages by utilizing traditional machine learning techniques combined with forensic feature extraction methods. These advantages highlight the practicality and efficiency of the approach used in this research.

One of the primary advantages of the proposed method is its computational efficiency. Unlike deep learning models that require large computational resources and specialized hardware such as GPUs, the proposed system can be implemented using standard computing systems. The combination of Error Level Analysis, statistical feature extraction, Principal Component Analysis, and Support Vector Machine classification allows the system to operate effectively without requiring high-end hardware.

Another important advantage is the simplicity and interpretability of the model. Traditional machine learning approaches rely on handcrafted features that are derived from observable characteristics of the images. In this research, Error Level Analysis highlights compression inconsistencies within images, while statistical features describe the distribution of pixel values. These features provide clear insights into the types of artifacts that may indicate image manipulation. This level of interpretability can be beneficial in digital forensic investigations where understanding the underlying evidence of manipulation is important.

The use of dimensionality reduction through Principal Component Analysis also improves the efficiency of the system. PCA reduces the number of features while retaining the most important information from the original feature space. This helps decrease computational complexity and improves the stability of the classifier during training.

Another advantage of the proposed system is the use of the Support Vector Machine classifier, which is well known for its effectiveness in high-dimensional classification tasks. The SVM algorithm can identify optimal decision boundaries between real and fake images even when

the dataset contains complex feature relationships. The use of the Radial Basis Function kernel further enhances the classifier's ability to model nonlinear patterns within the feature space.

Additionally, the proposed approach is relatively easy to implement and reproduce. The entire system can be developed using widely available open-source libraries in the Python programming language. This makes it accessible for researchers and practitioners who wish to explore deepfake detection without relying on large-scale deep learning frameworks.

6.5 Limitations of the Study

While the proposed deepfake detection system demonstrates the capability of identifying manipulated images using traditional machine learning techniques, several limitations should be acknowledged. Understanding these limitations is important for evaluating the scope of the study and identifying areas where improvements can be made in future research.

One of the primary limitations of this study is the moderate classification accuracy achieved by the model. Although the system successfully detects a significant portion of manipulated images, the overall accuracy of approximately 67% indicates that the classifier still makes a noticeable number of incorrect predictions. These errors occur mainly due to false positives, where real images are incorrectly classified as fake, and false negatives, where manipulated images are not detected by the system.

Another limitation arises from the use of handcrafted features. The proposed system relies on features extracted from Error Level Analysis and statistical measurements of pixel intensity distributions. While these features are capable of detecting certain compression artifacts and irregularities, they may not capture the complex patterns present in advanced deepfake images generated using modern artificial intelligence techniques. As a result, some subtle manipulations may remain undetected.

The size and diversity of the dataset used in this study may also influence the performance of the detection model. Machine learning algorithms generally require large and diverse datasets to learn robust patterns that generalize well to new data. If the dataset used in this research is limited in size or lacks sufficient variation in image conditions, the classifier may not perform as effectively when applied to new or unseen images.

Another limitation is the dependence on JPEG compression artifacts for detecting manipulation. Error Level Analysis is particularly effective when images have undergone

compression. However, if an image has been saved in a lossless format or processed in a way that minimizes compression artifacts, the effectiveness of ELA-based detection may decrease.

In addition, the proposed system focuses only on image-based deepfake detection. Deepfake technology is also widely used in videos, where manipulated frames are combined with audio and motion synthesis techniques. Detecting deepfake videos often requires additional analysis methods such as temporal consistency checks and frame-level feature extraction.

Finally, the proposed approach does not incorporate advanced deep learning techniques, which are currently widely used in state-of-the-art deepfake detection systems. Deep learning models have the ability to automatically learn complex features from raw image data and may achieve higher detection accuracy when trained on large datasets.

6.6 Future Work

The results of this research demonstrate that traditional machine learning techniques can be used to detect manipulated images to a certain extent. However, there are several opportunities to improve and extend the proposed deepfake detection system in future research.

One potential direction for future work is the use of larger and more diverse datasets. Machine learning models generally perform better when trained on datasets that contain a wide range of image variations, including different lighting conditions, resolutions, compression levels, and manipulation techniques. Expanding the dataset would allow the model to learn more robust patterns and improve its ability to generalize to unseen images.

Another important improvement would be the integration of additional feature extraction techniques. In this study, Error Level Analysis and statistical features were used to identify manipulation artifacts. Future research could incorporate other image forensic techniques such as texture analysis, frequency-domain features, or noise pattern analysis. Combining multiple feature extraction methods may help capture a wider range of manipulation characteristics and improve classification performance.

Future work could also explore the use of hybrid detection approaches that combine traditional machine learning with deep learning techniques. For example, deep learning models could be used to automatically extract complex image features, while traditional machine learning classifiers could be used for final decision-making. Such hybrid approaches may provide improved detection accuracy while maintaining interpretability.

Another possible extension of this research is the development of real-time deepfake detection systems. Implementing the proposed detection framework as part of an automated system could help identify manipulated images during online content verification, social media monitoring, or digital forensic investigations.

Additionally, future studies could extend the proposed framework to support video deepfake detection. Video deepfakes involve manipulation across multiple frames and may require analysis of temporal consistency and motion patterns. Incorporating frame-by-frame analysis and temporal feature extraction could improve the detection of manipulated video content.

Finally, improving the optimization and tuning of machine learning models may also enhance performance. Techniques such as hyperparameter optimization, ensemble learning, and advanced classification algorithms could be investigated to achieve better classification accuracy.

6.7 Practical Applications

The development of reliable deepfake detection systems has become increasingly important due to the rapid growth of artificial intelligence-based media generation technologies. Deepfakes have the potential to create realistic but manipulated images and videos that can mislead individuals and organizations. Therefore, systems capable of detecting such manipulations have several practical applications across different fields.

One important application of deepfake detection systems is in the field of digital forensics. Digital forensic investigators often analyze images and videos to determine whether media evidence has been manipulated. The proposed system can assist investigators by identifying compression artifacts and inconsistencies within images that may indicate tampering. Such tools can support forensic examinations in criminal investigations, cybercrime cases, and legal proceedings.

Another important application is in social media and online content verification. With the widespread use of social media platforms, manipulated images can easily spread misinformation and influence public opinion. Automated detection systems can help identify suspicious images before they are widely shared. This can assist platforms in reducing the spread of misleading or manipulated media content.

Deepfake detection systems can also play a role in journalism and media verification. News organizations frequently rely on images and videos as evidence when reporting events. Tools

capable of identifying manipulated media can help journalists verify the authenticity of visual content before publishing it, thereby improving the credibility of news reporting.

In addition, such systems may be useful in cybersecurity and identity protection. Deepfake technology can be used to create fake images that impersonate individuals. Detecting manipulated images can help prevent identity misuse, online fraud, and other forms of digital impersonation.

Educational and research institutions can also benefit from deepfake detection systems. These tools can be used as training resources for students and researchers studying digital forensics, cybersecurity, and artificial intelligence. By analyzing manipulated media, students can gain practical experience in identifying digital artifacts and understanding forensic analysis techniques.

CHAPTER 7

CONCLUSION

7.1 Summary of the Study

This research focused on the detection of manipulated images using traditional machine learning techniques. With the rapid advancement of artificial intelligence-based media generation technologies, deepfake images have become increasingly realistic and difficult to identify through visual inspection alone. This creates significant challenges in areas such as digital forensics, online media verification, and cybersecurity.

The main objective of this study was to develop a lightweight deepfake detection framework capable of distinguishing between real and fake images using forensic feature analysis and machine learning classification. The proposed system utilized Error Level Analysis (ELA) and statistical feature extraction techniques to capture compression inconsistencies and pixel distribution characteristics within images.

After feature extraction, a high-dimensional feature vector was constructed to represent each image. To improve computational efficiency and reduce redundancy in the feature space, Principal Component Analysis (PCA) was applied for dimensionality reduction. The reduced feature vectors were then used to train a Support Vector Machine (SVM) classifier with a Radial Basis Function kernel.

The experimental evaluation was conducted using a dataset containing both real and manipulated images. The dataset was divided into training and testing subsets using an 80:20 train-test split, and 5-fold cross-validation was applied to ensure reliable model evaluation. The performance of the proposed system was evaluated using multiple classification metrics including accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

The experimental results showed that the proposed system achieved an accuracy of approximately 67%, a precision of 63%, a recall of 80%, an F1-score of 70%, and an AUC score of 0.71. These results indicate that the system is capable of detecting manipulated images with moderate performance while successfully identifying a large portion of fake images.

7.2 Key Contributions of the Study

This research contributes to the growing field of deepfake detection by exploring the use of traditional machine learning techniques combined with digital forensic feature analysis. The study demonstrates that manipulated images can be detected by analyzing compression artifacts and statistical patterns present in digital images.

One of the key contributions of this study is the development of a lightweight deepfake detection framework that does not rely on complex deep learning architectures. Instead, the proposed system uses a combination of Error Level Analysis and statistical feature extraction techniques to identify potential manipulation artifacts within images. This approach highlights the usefulness of forensic-based image analysis for detecting digital image tampering.

Another important contribution of the research is the integration of dimensionality reduction using Principal Component Analysis. By reducing the high-dimensional feature space, PCA improves computational efficiency while preserving the most significant information required for classification. This step helps improve the stability of the machine learning model and reduces the risk of overfitting.

The use of a Support Vector Machine classifier with a Radial Basis Function kernel is another significant aspect of the proposed system. The SVM classifier effectively handles high-dimensional feature vectors and identifies nonlinear decision boundaries between real and fake images. This makes it a suitable choice for classification tasks involving complex feature patterns.

The study also contributes to the understanding of how forensic image analysis techniques can be combined with machine learning methods to detect manipulated media. The results demonstrate that traditional machine learning models can identify certain deepfake artifacts without requiring large-scale datasets or specialized hardware.

7.3 Limitations of the Research

Although the proposed deepfake image detection system demonstrates the ability to identify manipulated images using traditional machine learning techniques, several limitations should be acknowledged. These limitations highlight the constraints of the current study and provide context for interpreting the experimental results.

One of the primary limitations of this research is the moderate classification accuracy achieved by the model. The experimental results showed an overall accuracy of approximately 67%, which indicates that the classifier is able to detect manipulated images to a certain extent but still produces a noticeable number of incorrect predictions. These errors include both false positives, where real images are incorrectly classified as fake, and false negatives, where manipulated images are not detected.

Another limitation arises from the use of handcrafted feature extraction techniques. In this study, Error Level Analysis and statistical features were used to identify compression inconsistencies and pixel distribution patterns. While these features are useful for detecting certain types of image manipulation, they may not capture the full complexity of modern deepfake generation techniques. Deepfake images created using advanced artificial intelligence methods may contain subtle manipulations that are difficult to detect using traditional forensic features alone.

The size and diversity of the dataset used in this research also present a limitation. Machine learning models generally require large and diverse datasets to learn robust patterns and generalize effectively to unseen data. If the dataset used for training and testing is limited in size or does not include sufficient variations in image conditions, the model's ability to perform accurately on new data may be reduced.

Another limitation is the dependence on compression artifacts for detection. Error Level Analysis is particularly effective for images stored in compressed formats such as JPEG. However, if manipulated images are saved in formats with minimal compression artifacts or are carefully processed to remove such artifacts, the effectiveness of ELA-based detection may decrease.

Furthermore, the proposed system focuses only on image-based deepfake detection. Deepfake technology is widely used not only for images but also for videos, where manipulation occurs across multiple frames. Detecting deepfake videos requires additional analysis techniques such as temporal consistency analysis and motion pattern evaluation, which were not included in the scope of this study.

7.4 Recommendations for Future Research

The findings of this research demonstrate that traditional machine learning techniques can provide a practical approach for detecting manipulated images. However, several improvements and extensions can be explored in future studies to enhance the effectiveness and reliability of deepfake detection systems.

One potential direction for future research is the use of larger and more diverse datasets. Training machine learning models on a broader dataset containing various image conditions, manipulation techniques, and compression levels can improve the model's ability to generalize

to unseen images. A more diverse dataset would allow the classifier to learn more robust patterns and reduce misclassification errors.

Another important area for improvement is the incorporation of advanced feature extraction techniques. In this study, Error Level Analysis and statistical features were used to detect manipulation artifacts. Future research could explore additional forensic features such as texture analysis, frequency-domain features, or noise pattern analysis. Combining multiple types of features may provide a more comprehensive representation of image characteristics and improve classification performance.

Future studies may also investigate the use of deep learning models for deepfake detection. Deep learning architectures, such as convolutional neural networks, have demonstrated strong performance in image classification tasks by automatically learning hierarchical features from raw image data. Integrating deep learning techniques with traditional machine learning methods may lead to improved detection accuracy.

Another promising direction is the development of hybrid detection systems that combine traditional forensic analysis with deep learning approaches. For example, deep learning models could be used to extract complex features from images, while traditional machine learning classifiers could perform the final classification. Such hybrid systems may benefit from the strengths of both approaches.

Future research could also focus on real-time deepfake detection systems. Implementing detection algorithms in real-time applications would enable the automated identification of manipulated media on digital platforms, social media networks, or content verification systems. This would help reduce the spread of manipulated media and misinformation.

Additionally, extending the proposed framework to support video deepfake detection would be a valuable area of investigation. Video deepfakes involve manipulation across multiple frames and may require temporal analysis techniques to detect inconsistencies in motion, facial expressions, and frame transitions.

CHAPTER 8

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